ON THE INCLUSION $D_5(G) \subseteq \gamma_4(G)$: REDUCTIONS, THE LIE-RING ANALOGUE, AND COMPUTATIONAL RESULTS

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ABSTRACT. For a group G let $D_n(G) = G \cap (1 + \Delta(G)^n)$ be its integral dimension subgroups and $\gamma_n(G)$ its lower central series. Whether $D_5(G) \subseteq \gamma_4(G)$ for every group G is open (Kourovka Notebook, Problem 20.84); for Lie rings the analogous inclusion was recently proved by Ionin, Mikhailov and Petrov and formally verified in Lean. We reduce the group question, by an elementary argument, to finite metabelian 2-groups of class three, and identify the steps of the Lie-ring proof that have no group-theoretic counterpart. A computer search shows that a counterexample among finite 2-groups of class three has order at least 2^{12}. For 3-torsion in D_5/γ_5 we prove that $D_5(G) = \gamma_5(G)$ whenever $G/\gamma_2(G)$ is a finite 3-group with at most one invariant larger than 3, and, with the help of a computer search, for all 3-groups of order at most 3^9. Along the way we give an exact criterion for a p-group of type $(d, 2, 1)$ with elementary abelian $P/\gamma_2(P)$, p odd, to have immediate descendants, and an explicit graded Lie algebra of type $(3, 3, 1, 1)$ over $\mathbb{F}_p$ with $[L_2, L_2] \neq 0$; it shows that the Lie-ring lemma behind our metabelian results requires γ_2/γ_3 to have at most two generators.

Conventions on proofs. Results whose proofs depend on computer calculations are labelled or noted as *computer-assisted*: Propositions 5.1, 5.3 and 5.10, Theorems 5.4 and 6.14, and the first statement of Proposition 5.7. Other computational facts, such as examples, counts and checks, are identified as computations where they occur. All other proofs are by hand: here, in the cited sources, or, where indicated, in the documentation accompanying the code. The computations use GAP [GAP] with the SmallGroups library [BEO02] and the ANUPQ package [GNO], together with our own code; §5 explains why the lattice computations modulo a prime power are exact and how their results were checked by independent implementations. The code, raw tables and checks are available at `https://github.com/ispacex/dimension-subrings`. Many of the results proved by hand are, in addition, formally verified in Lean 4 with Mathlib; the formalization is available at `https://github.com/ispacex/group-dimension`. This is noted at each such result, and §4 states the formal versions with their exact

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hypotheses. For the 3400 groups of Proposition 5.3 the formalization also contains certificates of $D_5 = 1$, checked with `native_decide`.

1. The question

Let $\Delta = \Delta(G)$ be the augmentation ideal of $\mathbb{Z}[G]$ and $D_n(G) = \{g \in G : g - 1 \in \Delta^n\}$. Always $\gamma_n(G) \subseteq D_n(G)$ (this is also formally verified, §4), with equality for $n \leq 3$ and failure for $n = 4$ in general (Rips [Rip72]).

Question 1.1 ([KM26, Problem 20.84], [IMP26, §1.3])**.** *Is $D_5(G) \subseteq \gamma_4(G)$ for every group G?*

Known facts used below, all from the cited sources (the first one is also formally verified, see §4): $D_4(G)^2 \subseteq \gamma_4(G)$ (Passi [Pas68], see also [MP09]); $D_n(G)/\gamma_n(G)$ is abelian [GK92]; the exponent of $D_5(G)/\gamma_5(G)$ divides $3! = 6$ (Tahara [Tah81, Cor. 6.11], proved there for finite groups, to which the general case reduces by the standard argument; Tahara also gives an explicit description of D_5 for finite groups of class 4 [Tah81, Thm. 6.1] and shows $D_k = \gamma_k$ for $k \leq 5$ when G has no elements of order 2 or 3 [Tah81, Cor. 6.12]); Gupta's claim that all D_n/γ_n are 2-groups [Gup02] is false: Bartholdi and Mikhailov constructed a 3-group with $D_7 \neq \gamma_7$ [BM23].

Related work. For metabelian groups much more is known. Gupta proved $D_n(G) = \gamma_n(G)$ for $n \leq 2p - 1$ in metabelian p-groups [Gup88], and that for a finite metabelian group G the exponent of $D_n(G)/\gamma_n(G)$ is a power of 2; in particular $D_n = \gamma_n$ for all n in finitely generated metabelian p-groups with p odd [Gup91]. This result is taken as proved in [BM23, §1.1], unlike his later claims for all groups [Gup02]. Sjogren bounded the exponent of $D_n(G)/\gamma_n(G)$ by a function of n alone [Sjo79]. Question 1.1 is also Problem 16.66(a) of [KM26] (Mikhailov and Passi): it asks for $f(3)$, the largest m with $D_m(G) \neq 1$ for some nilpotent group G of class 3, and by Rips's example the question is whether $f(3) = 4$. Examples of 2-groups of class 3 with $D_4 \neq 1$ and $D_5 = 1$ are in [SGu16]; conditions under which $D_5 = \gamma_5$ for certain finite 2-groups (of class 3 in [GK23]) and 3-groups, derived from Tahara's Theorem 6.1, are in [GK19, GK21, GK23]. We have not verified the arguments of [SGu16, GK19, GK21, GK23], and do not use their results.

Results and plan of the paper. Section 2 reduces Question 1.1 to finite metabelian 2-groups of class exactly 3 (Proposition 2.1). Section 3 summarises the proof of the Lie-ring analogue in [IMP26] and lists the steps that have no counterpart for groups; Section 4 describes the Lean formalization. Section 5 reports the computations: $D_5(Q) = 1$ for every 2-group Q of class 3 and order 2^{10} or 2^{11} (Propositions 5.3 and 5.10), so that a counterexample among finite 2-groups of class 3 has order at least 2^{12}, together with consequences for finite groups of small order (Theorem 5.4). Section 6 treats 3-torsion in D_5/γ_5: $D_5(G) = \gamma_5(G)$ whenever $G/\gamma_2(G)$ is a finite 3-group with at most one invariant larger than 3 (Theorems 6.6 and 6.8), and for every 3-group of order at most 3^9 (Theorem 6.14); on the way it proves the criterion for immediate descendants of Theorem 6.12. Section 7 collects

open questions. Appendix A gives the Lie algebra of type $(3,3,1,1)$, and Appendices B and C contain the longer proofs.

2. An elementary reduction

Proposition 2.1. *The following are equivalent.*

(1) $D_5(G) \subseteq \gamma_4(G)$ *for every group* G.
(2) $D_5(M) = 1$ *for every metabelian group* M *of nilpotency class at most* 3.
(3) $D_5(Q) = 1$ *for every finite metabelian* 2*-group* Q *of nilpotency class exactly* 3.

Proof. (The proposition is also formally verified, without hypotheses, since Passi's and Gruenberg's theorems and $D_3 = \gamma_3$ are formalized too, and with "metabelian" dropped by Remark 2.2; see §4.)

(1) $\Rightarrow$ (3) is clear since $\gamma_4(Q) = 1$.

(2) $\Rightarrow$ (1). Put $N = \gamma_4(G)G''$. Since $G'' = [\gamma_2(G), \gamma_2(G)] \subseteq \gamma_4(G)$, $N = \gamma_4(G)$, and $M = G/N$ is metabelian of class ≤ 3. The projection $\pi\colon G \to M$ induces a ring map $\mathbb{Z}[G] \to \mathbb{Z}[M]$ sending $\Delta(G)^n$ into $\Delta(M)^n$, so $\pi(D_5(G)) \subseteq D_5(M) = 1$, i.e. $D_5(G) \subseteq N = \gamma_4(G)$. (This is the group version of the step from Theorem C to Corollary D in [IMP26].)

(3) $\Rightarrow$ (2). Let M be metabelian of class ≤ 3 and $1 \neq x \in D_5(M)$. *Finite generation.* $x - 1$ is a finite $\mathbb{Z}$-linear combination of products $(g_1 - 1)\cdots(g_5 - 1)$; all g_i and x lie in a finitely generated subgroup H, and then $x \in D_5(H)$. H is metabelian of class ≤ 3, hence finitely generated nilpotent. *Exponent two.* By Passi, $D_4(H)^2 \subseteq \gamma_4(H) = 1$; as $D_5 \subseteq D_4$, $x^2 = 1$. *Finite* 2*-quotients.* Let T be the (finite, normal) subgroup of elements of odd order of H; $x \notin T$. The group H/T is finitely generated nilpotent with 2-group torsion, hence residually a finite 2-group (Gruenberg [Gru57]). Choose a finite 2-group quotient Q of H in which the image of x is nontrivial; by functoriality it lies in $D_5(Q)$. Q is metabelian of class ≤ 3, and class ≤ 2 is impossible because then $D_5(Q) \subseteq D_3(Q) = \gamma_3(Q) = 1$. □

Remark 2.2 ("metabelian" is automatic). Every group of nilpotency class ≤ 3 is metabelian: $G'' = [\gamma_2(G), \gamma_2(G)] \subseteq \gamma_4(G)$ by P. Hall's inclusion $[\gamma_i, \gamma_j] \subseteq \gamma_{i+j}$ (see e.g. [MP09]); this is the same fact used in (2) $\Rightarrow$ (1). So in (2) and (3) the word "metabelian" may be dropped: (3) is the statement that $D_5(Q) = 1$ for every finite 2-group of class 3, and the search space of §5 is simply the 2-groups of class 3. The same holds for Lie rings ($[\gamma_2, \gamma_2] \subseteq \gamma_4$), so Theorem C of [IMP26] for $c = 3$ needs no metabelian hypothesis either; the hypothesis matters only for $c \geq 4$, where $D_{2c-1} = 1$ for all groups of class $\leq c$ is a different, stronger question. Accordingly, ANUPQ's option `Metabelian` cut nothing in our searches. That groups of class ≤ 3 are metabelian is also formally verified (§4).

Remark 2.3 (a bounded form). For finite groups the proof gives more. If G is finite then so are $M = G/\gamma_4(G)$ and $H \leq M$, and one may take

$Q = H/T$, a finite 2-group whose order is the 2-part of $|H|$, hence divides the 2-part of $|G|$. Therefore: *if $D_5(Q) = 1$ for all metabelian 2-groups Q of class 3 and order $\le 2^k$, then $D_5(G) \subseteq \gamma_4(G)$ for every finite group G whose Sylow 2-subgroup has order $\le 2^k$.* This is also formally verified (§4), without hypotheses and both for Q of class ≤ 3 and of class exactly 3, since Passi's theorem and $D_3 = \gamma_3$ are formalized; the formal proof does not need Gruenberg's theorem, as there $H = M$ and Q is the Sylow 2-subgroup of M, a direct factor since M is nilpotent. Note that the exponent step uses Passi's theorem for H (where $\gamma_4(H) = 1$), not for an arbitrary group.

Remark 2.4. Proposition 2.1(2) is precisely the group analogue of Theorem C of [IMP26] for $c = 3$ ("a metabelian Lie ring of class $\le c$ has $D_{2c-1} = 0$"). The reduction is elementary and almost certainly known to experts; we record it because it fixes the search space for computations (§5).

3. The Lie-ring proof and where it does not transfer

We summarise the proof of [IMP26, Theorem C, Corollary D], checked against its text; the final statements are formally verified in the Lean formalization accompanying [IMP26].

(1) Reduction to L metabelian of class $\le c$ (Corollary 6.14) and to finitely generated L (Lemma 6.13). *Transfers*: this is Proposition 2.1.
(2) A central extension $0 \to K \to T \to L \to 0$, $T = L \times_{L_{ab}} P$ with P free abelian, such that $\gamma_i(T) \cong \gamma_i(L)$ for $i \ge 2$ and $T \cong T' \oplus P$ additively (Lemma 6.4, Corollary 6.7).
(3) The integral PBW theorem: $\operatorname{gr} U(T) \cong \mathrm{Sym}_{\mathbb{Z}}(T)$ and, for T a direct sum of cyclic groups, an explicit $\mathbb{Z}$-basis of ordered monomials with known additive orders (Theorem 2.3, Corollary 2.4).
(4) The additive projection $p\colon U(T) \to P_{\le 2}U(T)$ onto monomials of length ≤ 2 (§6.2), and a retraction $r\colon P_{\le 2}U(T) \to T$ killing $Z(T) \cdot T$ (Lemma 5.3).
(5) The quadratic part of $p(\varpi(T)^n)$ lies in $\sum_{i+j=n} \gamma_i(T)\gamma_j(T)$ (Proposition 6.12); for $n = 2c-1$ every such term has a central factor, and Theorem 5.1 gives $D_{2c-1}(L) = 0$.

Obstacles for groups. The following is our analysis; where no source is cited, it is an explanation, not a theorem.

- *No PBW basis of the required kind.* Step 3 is the heart of the argument: it gives an additive splitting of $U(T)$ by monomial length, compatible with additive orders, so that "the quadratic part" of an element is well defined and computable. For $\mathbb{Z}[G]$ the filtration by powers of Δ has associated graded ring $\bigoplus \Delta^n/\Delta^{n+1}$, which in general is *not* the integral enveloping algebra of $\operatorname{gr} \gamma_\bullet(G)$; over $\mathbb{Q}$ the analogous comparison is Quillen's theorem [Qui68], and over $\mathbb{Z}$ its failure is essentially the dimension subgroup problem itself

(cf. [HMP07, MP09]). There is no canonical additive projection of $\mathbb{Z}[G]$ onto "length ≤ 2".

- *The extension T.* Step 2 has a formal group analogue (pull back a free abelian cover of G_{ab} along $G \to G_{ab}$, giving a central extension with the same higher lower central terms). What is missing is how to use it: in the Lie case T' is an additive direct summand and its elements commute in $U(T)$, which drives the rewriting Lemmas 6.9–6.10. In $\mathbb{Z}[G]$ products of commutators do not commute and the corresponding rewriting produces higher-order terms ("collection").
- *Torsion is already there in degree four.* For Lie rings $D_n = \gamma_n$ for $n \leq 3$ and $2D_4 \subseteq \gamma_4$ [BP15], the same pattern as for groups; but for groups even the description of D_4/γ_4 requires the full Rips–Passi machinery, and the known group computations of D_5 (e.g. the group of [IMP26, §3.3], where $D_5(G) = \langle w \rangle \subseteq \gamma_4(G)$) are case-by-case.

4. Formalization

The Lean 4 project `group-dimension` (Lean and Mathlib `v4.35.0`, with the module system; the earlier version on Mathlib `v4.29.0`, the one used by [IMP26], is kept as the tag `lean-v4.29`) defines $\Delta(G)$ as the kernel of the augmentation of `MonoidAlgebra` $\mathbb{Z}$ G and $D_n(G) = \{g : g - 1 \in \Delta^n\}$, with literature indexing; Mathlib's `(⊤ : Subgroup G).lowerCentralSeries k` (formerly `lowerCentralSeries G k`) is γ_{k+1}. The following are proved, with no `sorry` and only the standard axioms `propext`, `Classical.choice`, `Quot.sound`, except the certificates for Proposition 5.3, which also use `native_decide` (stated there):

- $D_n(G)$ is a normal subgroup, $D_0 = D_1 = G$, the series is decreasing, and $\Delta^i\Delta^j \subseteq \Delta^{i+j}$;
- the commutator identities $[a,b] - 1 = a^{-1}b^{-1}\big((a-1)(b-1) - (b-1)(a-1)\big)$ (for $[a,b] = a^{-1}b^{-1}ab$) and its analogue for Mathlib's convention; consequently $[D_i, D_j] \subseteq D_{i+j}$ and $\gamma_n(G) \subseteq D_n(G)$;
- functoriality of D_n under group homomorphisms;
- $D_2(G) = \gamma_2(G)$, via a ring map $\mathbb{Z}[G] \to \mathbb{Z} \oplus G_{ab}$ (square-zero extension) killing Δ^2;
- $D_3(G) = \gamma_3(G)$ for every group (Higman, Rees, Passi; `dThreeEq GammaThree`, file `DThree.lean`). For finite groups it follows from Tahara's Theorem 4.1, formalized below, applied to $G/\gamma_3(G)$. For an arbitrary group, an element of D_3 lies in D_3 of a finitely generated subgroup of $G/\gamma_3(G)$, and in a finitely generated group of class ≤ 2 every $c \neq 1$ of γ_2 survives in a finite quotient (by the structure theorem for finitely generated abelian groups and the central subgroup of m-th powers);
- Proposition 2.1 and Remark 2.3: first, $D_5(G) \subseteq \gamma_4(G)$ for all G iff $D_5(Q) = 1$ for all Q of class ≤ 3; then, with Passi's $D_4^2 \subseteq \gamma_4$ and Gruenberg's theorem as explicit hypotheses (both are proved

and discharged in the primed versions, see the next two items), iff $D_5(Q) = 1$ for all finite 2-groups of class ≤ 3, and (with $D_3 = \gamma_3$, proved above) also of class exactly 3; the bounded form for finite G does not need Gruenberg's theorem; "metabelian" is dropped by Remark 2.2, which is proved too (`metabelian_of_class_three`, file `Descendants.lean`; `dimensionQuestion5_iff_finite_two_groups`, `dimensionQuestion5_iff_class_three`, `D_five_le_of_bounded`, `D_five_le_of_bounded_class_three`; file `FiniteReduction.lean`). Finite generation is proved: every element of $\Delta(G)^n$ comes from $\Delta(H)^n$ for a finitely generated $H \leq G$;

- Gruenberg's theorem [Gru57] in the form used in Proposition 2.1: in a finitely generated nilpotent group an element none of whose odd powers is trivial survives in some finite 2-group quotient (`gruenbergResidualTwo`). Along the way, finitely generated nilpotent groups are residually finite (`Gruenberg.residuallyFinite`), by induction on the class c: the last term $A = \gamma_c(G)$ is central and finitely generated; for $g \neq 1$ choose $B \leq A$ of finite index with $g \notin B$ and $U \leq G$ of finite index with $[\gamma_{c-1}(G), U] \subseteq B$, so that $\gamma_c(U) \subseteq B$ and U/B has smaller class. Gruenberg's theorem is again by induction on the class: either the image of the element modulo $\gamma_c(G)$ has no trivial odd power and the induction hypothesis applies, or some odd power h of it is central with no trivial odd power; then $h \notin \langle h^2 \rangle$, residual finiteness of $G/\langle h^2 \rangle$ gives a finite nilpotent quotient in which h has order 2, and the projection onto its Sylow 2-subgroup keeps h, hence the element. File `Gruenberg.lean`;
- Passi's theorem [Pas68]: $D_4(G)^2 \subseteq \gamma_4(G)$ for every group (`passiDFourSquare`). This is not Passi's original argument: it goes through Tahara's map ψ_3 of [Tah81, Thm. 5.1], already formalized for his Theorem 6.1 (see below). For a finite p-group of class ≤ 3 take Tahara's frame (uniqueness bases of the layers with divisibility chains). For $g \in D_4$ the element $g - 1$ is a combination of proper products of weight ≥ 4, whose images have no component in W_3^*; so the W_3^*-part of the symbol of g is that of a combination of Tahara's relations. Its coordinates give Tahara's conditions (A_2), (A_3), (B) and $g = \prod_{i<j} \kappa_{ij}^{u_{ij}q_{ij}}$ with $\kappa_{ij} = [x_i^{d(i)}, x_j]$, $q_{ij} = d(j)/d(i)$. In the abelian group γ_3, written additively, the identity $[x^n, y] = [x, y]^n [[x, y], x]^{\binom{n}{2}}$ and (A_2), (A_3) give $u_{ij}q_{ij}\kappa_{ij} = -u_{ij}\kappa_{ji}$, and (B) gives $\sum_{i<j} u_{ij}\kappa_{ji} = \sum_{i<j} u_{ij}q_{ij}\kappa_{ij}$; hence $g = -g$, i.e. $g^2 = 1$. A finite nilpotent group reduces to its Sylow subgroups, and an arbitrary group, through $G/\gamma_4(G)$, finite generation and the residual finiteness of finitely generated nilpotent groups (previous item), to finite ones. Files `Passi.lean`, `TaharaProof/DFour.lean`. Hence Proposition 2.1

and Remark 2.3 without hypotheses (`dimensionQuestion5_iff_finite_two_groups'`, `dimensionQuestion5_iff_class_three'`, `D_five_le_of_bounded'`, `D_five_le_of_bounded_class_three'`; file `DThreeCorollaries.lean`);

- Lemma 5.6 and Proposition 5.7 for candidates: no candidate has $|P_2(Q)| \leq 8$, so there is none of order 2^{11} and rank ≥ 8 (`false_of_card_expCentral_two_le`, `not_isCandidate_of_rank`, `not_isCandidate_of_quotient_z4`; file `Candidate.lean`). Here $d(Q)$ is the least number of generators, and M_n is defined by the formula $\prod_{ip^j \geq n} \gamma_i^{p^j}$; that it equals $Q \cap (1 + \Delta_{\mathbb{F}_2}(Q)^n)$ is Jennings' theorem, and the inclusion needed here is proved: $D_5(Q) \subseteq Q \cap (1 + \Delta_{\mathbb{F}_2}(Q)^5) \subseteq Q^8\gamma_2^4\gamma_3^2$ for finite 2-groups of class ≤ 3 (`D_five_le_jennings`). The proof is Jennings': for a finite p-group, $Q \cap (1 + \Delta^n_{\mathbb{F}_p})$ lies in the n-th term of every restricted N-series, via the basis $\prod(x_i - 1)^{a_i}$, $0 \leq a_i < p$, of $\mathbb{F}_p[Q]$; for the smallest one, the Lazard series, and $p = 2$, $n = 5$, class ≤ 3 its fifth term is checked to lie in $Q^8\gamma_2^4\gamma_3^2$ (files `Jennings/`). That the Lazard series equals $\prod \gamma_i^{p^j}$ in general (Lazard) is not formalized. Lemma 5.2(a)–(c) is proved (`isCandidate_of_minimal`, with $D_3 = \gamma_3$ as a hypothesis, discharged in `isCandidate_of_minimal'`, file `DThreeCorollaries.lean`), and so is (d) (`isDescendant_of_minimal`, file `Descendants.lean`): $P_c(Q) = \langle n \rangle$ has order 2, and $R = Q/\langle n \rangle$ has $\gamma_4(R) = 1 \neq \gamma_2(R)$, $P_c(R) = 1 \neq P_{c-1}(R)$ and $\gamma_3(R)$ cyclic. The D_5 statement of Proposition 5.7 is proved with Proposition 5.3 as an explicit hypothesis (the hypothesis $D_3 = \gamma_3$ is discharged in `D_five_eq_bot_of_rank'`, file `DThreeCorollaries.lean`; with class ≤ 3 in place of 3, here and in the hypothesis at order 2^{10}, which given $D_3 = \gamma_3$ is Proposition 5.3; the latter is a computation; `D_five_eq_bot_of_rank`, file `MinimalCounterexample.lean`). Also proved: P_2 lies in every maximal subgroup of a finite p-group, so $d(G) \leq n$ when $|G/P_2| \leq p^n$;
- Lemma 5.9 (adjoining a central root): $\mathbb{Z}[H] \cap \Delta(K)^n = \Delta(H)^n$ and $D_n(K) = D_n(H)$ for $n \geq 2$ (`D_n_eq_of_index`, file `CentralRoot.lean`);
- Proposition 5.8, all cases, with the ring statement $\mathbb{Z}[H] \cap \Delta(Q)^n = \Delta(H)^n$, for $Q = H\langle a, b \rangle$ with a, b centralising H and $a, b, ab \notin H$; also the step of adjoining a central square root in (2) and the variant $c^{2^j} \in \gamma_{j+2}(H)$ (`D_n_eq_of_dihedral`, `D_n_eq_of_sqrt`, `D_n_eq_of_central_product` and its corollaries; files `CentralD8.lean`, `CentralSqrt.lean`, `CentralProduct.lean`). The passage from $(H \times E)/\{(\varphi(z), z^{-1})\}$ to this form and the classification of E are not formalized. The formal proof uses the same overgroup $(Q \times \mathbb{Z})/\langle (s, -2) \rangle$ for Q_8 and $M_{2^{k+2}}$, with $s = a^2$ ($s = t$ for Q_8) and $b^2 = s^e$ ($e = 1$ for Q_8, $e = 0$ for $M_{2^{k+2}}$): the image ζ of the

generator of $\mathbb{Z}$ has $\zeta^2 = s$, and $a\zeta^{-1}$, $b\zeta^{-e}$ generate a subgroup D_8 centralising $H\langle\zeta\rangle$;

- Lemma 6.1, with Gupta–Tahara's $D_{p+2} = \gamma_{p+2}$ for metabelian p-groups, p odd [GT85, Thm. 3.2(c)], as an explicit hypothesis, stated for finite p-groups (`D_le_derivedSeries_two`, `mem_of_mem_D_add_two`, `rank_ge_three_of_D_five_ne_bot`; file `MetabelianOdd.lean`); here $M_n(Q) = Q \cap (1 + \Delta_{\mathbb{F}_p}(Q)^n)$, whose equality with $\prod_{ip^j \geq n} \gamma_i^{p^j}$ (Jennings) is formalized only in the inclusion needed for $p = 2$, $n = 5$;
- Lemma 6.2, with Gupta–Tahara's theorem as an explicit hypothesis (`minimal_three'`, file `DThreeCorollaries.lean`; Passi's theorem, a hypothesis of `minimal_three`, is proved), "least order" meaning that $D_5(R) = \gamma_5(R)$ for every 3-group R with $|R| < |Q|$, and $\Omega_1(Z(Q)) = \langle z\rangle$ stated as: every central x with $x^3 = 1$ lies in $\langle z\rangle$ (`minimal_three`; the form with hypotheses on the quotients by central subgroups of order 3, as in Lemma 5.2, is `minimal_three_of_quotients`; file `MinimalThree.lean`). Also proved: the 3-class c is ≥ 4, and $P_c(Q/\langle z\rangle) = 1 \neq P_{c-1}(Q/\langle z\rangle)$; a finite abelian p-group with only one subgroup of order p is cyclic;
- Lemma 6.3 and Proposition 6.5, with $d(Q)$ the least number of generators (`commutator_gamma_two_le_gamma_five`, `metabelian_of_class_four`, `card_ge_of_not_metabelian`, `metabelian_of_card_le`, `rank_of_card_nine`, `type_of_card_nine`; files `ClassFour.lean`, `ClassFourOrder.lean`); in the proposition the class is assumed ≤ 4 instead of $= 4$. Instead of the Lie ring $L(G)$ the proof works in $G/\gamma_5(G)$, where γ_4 is central: the brackets become $F([a,b])$ for an additive $F\colon \gamma_4 \to \mathbb{F}_p$, nonzero at a given commutator (it exists since $\gamma_4(G/\gamma_5)$ has exponent p), and both Jacobi identities come from the Hall–Witt identity. Also proved on the way: $[\gamma_i, \gamma_j] \subseteq \gamma_{i+j}$, a group of order p^n is generated by n elements, and in a finite p-group $H\gamma_2 = G$ implies $H = G$. That both types occur for $p = 3$ is a computation and is not formalized;
- the algebra of Appendix A (Remark 6.4(2)), with $[e, \rho] = 2t$: over any commutative ring R it is a Lie algebra generated by e, k_1, k_2, and γ_k is spanned by the basis elements of degree $\geq k$; over any field $\dim \gamma_1, \dots, \gamma_5 = 8, 5, 2, 1, 0$, i.e. the type is $(3, 3, 1, 1)$, and $[\gamma_2, \gamma_2] \ni w \neq 0$ (`lieSpan_eq_top`, `lcs_eq`, `finrank_lcs`, `lie_lcs_one_ne_bot`, `exists_lie_three_three`; file `LieThreeThree.lean`). The bracket is given in coordinates on R^8, so the Jacobi identity is a polynomial identity;
- Theorem 6.10, for every group Q with a central subgroup M of exponent p and $Q/M \cong P$ (Mathlib has no p-covering group; P^* with its multiplier is such a group), and for M of order p, which covers the immediate descendants of step size 1 (`metabelian_`

`of_central_elementary`, `metabelian_of_central_quotient`; file `PCover.lean`); P is assumed of class ≤ 3 and not necessarily finite. The proof is the paper's, with its first two cases merged: Q'' lies in $\langle z\rangle \cap \gamma_4(Q)$, which is trivial unless $z \in \gamma_4(Q)$;

- Lemma 6.9, for $Q = E/U$ with $U \leq M$ subgroups of any group E, U normal, $M \not\leq U$ and the nucleus replaced by any subgroup $N \not\leq U$ (`filter_on_parents`; file `Descendants.lean`); M_5 is the product $\prod_{ip^j \geq 5} \gamma_i^{p^j}$, and the condition on $\gamma_4(P^*)$ is proved without assuming that P has class 3. On the way: γ_n, the derived series, P_n and $\prod_{ip^j \geq n} \gamma_i^{p^j}$ commute with surjective homomorphisms;
- Theorem 6.12, the implication "⇒" (Steps 2 and 4 of Appendix C), with "nonterminal" in the form of Lemma C.1: there is a group Q with a central subgroup $\langle z\rangle$ of order p, $Q/\langle z\rangle \cong P$ and $\gamma_4(Q) \neq 1$ (`nonterminal_necessary`; file `Nonterminal.lean`); P need not be finite. (N4) is proved for every choice of u, v, w and of any finite family $x_1, \ldots, x_n$, which need not generate P, and of the exponents. As in the proof of Lemma 6.3, Step 2 is carried out in the group: $[[\ell, x], y] = [[\ell, y], x]$ for $\ell \in \gamma_2(Q)$, with $Q'' = 1$ from Theorem 6.10. On the way: $[x, y, t][y, t, x][t, x, y] = 1$ in every metabelian group, and $(xa)^n = x^n a^n b^{\binom{n}{2}} c^{\binom{n}{3}}$ for $b = [a, x]$, $c = [b, x]$ with a, b, c commuting and $[c, x] = 1$, as in Step 4. Lemma C.1 (that this condition is equivalent to having an immediate descendant, i.e. a nontrivial nucleus [OBr90]) is not formalized;
- Theorem 6.12, the implication "⇐" (Step 5 of Appendix C), in the same form, assuming of P only $\gamma_4(P) = 1$ and $|\gamma_3(P)| = p$ besides (N2), for P generated by $x_1, \ldots, x_d$ with $|P/\gamma_2(P)| = p^d$ (`nonterminal_sufficient`; files `NonterminalSufficient.lean`, `IterExt.lean`); P need not be finite. Together with the previous item: for a given normal form as in Step 3, P is nonterminal in the form of Lemma C.1 if and only if (N2) holds and (E)+(J) has a solution with $c = 1$ (`nonterminal_iff`). The group Q is built differently from the paper, without a pc-presentation and its consistency test: starting from the abelian group $A = \langle u, v, w, z\rangle$ with $u^p = z^{\mu_u}$, $v^p = z^{\mu_v}$, $w^p = z^p = 1$ (a quotient of $\mathbb{Z}^4$), the generators $x_1, \ldots, x_d$ are adjoined one at a time as semidirect products with $\mathbb{Z}$, which needs only that conjugation by the new generator is an automorphism of the group built so far; this is the Jacobi identity, whose z-component is (J) and whose remaining part holds in P. The elements $x_i^p v^{-\alpha_i} w^{-\beta_i}$ of this group are central: they commute with A as $(1 + D_{x_i})^p = 1$, and with the x_j by (E). Q is the quotient by the subgroup they generate, and its kernel onto P is $\langle z\rangle$: z has order p in Q via a homomorphism to $\mathbb{Z}^d$, and the kernel lies in $\langle z\rangle$ by the independence of the x_i in $P/\gamma_2(P)$. Here (and in `nonterminal_iff`) P and Q are in Lean's `Type`, while "⇒"

allows any universe; in both directions the normal form (Step 3) is given, not constructed;

- Tahara's Theorem 6.1 [Tah81], the inclusion: for a finite group of class ≤ 4 and any uniqueness bases (with no basis element in the next term of the series) with their structure constants, D_5 lies in the subgroup generated by the elements g over the integer families satisfying (A)–(F); the bases, structure constants, conditions and generators are written out as in [Tah81, pp. 143, 150, 159–161] (`taharaFiveOne` proves `TaharaFiveOne`; file `TaharaTheorem.lean`, proof in `TaharaProof/`). The proof follows §§2–6 of [Tah81]: the uniqueness bases of the layers are glued into one ordered family with normal forms, the basic products form a $\mathbb{Z}$-basis of $\mathbb{Z}[G]$, and Δ^n is spanned by the products of weight $\geq n$ modulo lower ones; hence $D_3 = \gamma_3$ for these groups (Tahara's Theorem 4.1), and the map ψ_3 of Tahara's Theorem 5.1 into $W_3^* \oplus W_4$, with Losey's basis, sends Δ^5 into the span R_5^* of the relations (I)–(VII). For $g \in D_5$ the coordinates of $\psi_3(g-1)$ give (A)–(F), and its $A_3 \oplus A_4$ part identifies g with a generator. The Lean statement of condition (C_2) was checked against [Tah81, p. 160];
- Theorems 6.6 and 6.8, with no hypotheses (`D_five_eq_gamma_five_of_exp_three`, `D_five_eq_gamma_five_of_cubes_cyclic`; file `TaharaTheorem.lean`). They are proved with Tahara's Theorem 6.1 as an explicit hypothesis (`D_five_eq_gamma_five`, `D_five_eq_gamma_five_cyclic`; files `ExpThree.lean`, `ExpThreePlus.lean`), and the hypothesis is discharged by the previous item. In Theorem 6.8 the hypothesis reads "G/γ_2 is a finite 3-group whose cubes form a cyclic group", which for a finite abelian 3-group means at most one invariant > 3. The reduction to the finite group G/γ_5 is proved too. Elements of γ_3 and γ_4 are compared through additive maps to $\mathbb{F}_3$, and the uniqueness bases adapted to the cube map come from bases of $\mathbb{F}_3$-spaces;
- Proposition 5.3 for the 3400 presented groups, by certificates (see the paragraph after the proof of Proposition 5.3): $D_4 = D_5 = 1$ and order 2^{10} for each (`D_four_cand`, `D_five_cand`, `card_cand`). `prop_1024` derives Proposition 5.3 from $D_3 = \gamma_3$ (discharged in `prop_1024'` and `D_five_eq_bot_of_rank_of_complete'`), Proposition 5.1 at order 2^9 (with Remark 2.2, for all groups of class ≤ 3) and the completeness of the list (every candidate of order 2^{10} is isomorphic to one of the 3400; this is Lemma 5.2(d), formally verified, together with the ANUPQ search, a computation that is not formalized), all three as explicit hypotheses; `D_five_eq_bot_of_rank_of_complete` is the D_5 statement of Proposition 5.7 with the same hypotheses. All of these use `native_decide`: besides the three standard axioms they rest on the 34 axioms it introduces, i.e. they trust the Lean compiler that evaluated the checks, not only the kernel;

- in the version `lean-v4.29` only: the Lie-ring inclusion $D_5(L) \subseteq \gamma_4(L)$, derived from the authors' `LieRings.Publication.corollaryD` [IMP26]. The current version does not import the authors' package (it has no licence file and is on Mathlib `v4.29.0`) and only quotes that statement.

Stated as named propositions with references, *not* proved: Question 1.1; Rips's example; Gupta–Tahara's [GT85, Thm. 3.2(c)] for finite p-groups; Tahara's bound; the two Gupta–Kuz'min statements of [GK92]; $D_5^2 \subseteq G''\gamma_5$ [MP09, Thm. 2.28]; and the group analogues of Theorem C and Corollary D of [IMP26] (without any claim about their truth). Some of them are explicit hypotheses of theorems (as Gupta–Tahara's theorem in Lemma 6.2); none is assumed globally.

Nine of the formal results above are registered in the Palomar registry of formalized results (`https://palomar-registry.org`) as PALOMAR-2026-10-23-000007, after passing its mechanical check: Theorems 6.6 and 6.8; Lemma 5.6; Proposition 5.7 for candidates (two statements: no candidate has $|P_2(Q)| \leq 8$, and there is none of order 2^{11} and rank ≥ 8); Jennings' inclusion; and, with their hypotheses explicit, Proposition 2.1 ((1)⇔(3), in the registered version with Gruenberg's and Passi's theorems as hypotheses), Lemma 5.2(a)–(c) and the D_5 statement of Proposition 5.7.

5. Computations

All code, raw tables and the checks are in `compute/` of the repository `https://github.com/ispacex/dimension-subrings` (`README.md`, `results/SUMMARY.md`).

Method. For finite G, Δ is free abelian with basis $\{g-1\}_{g\neq 1}$, and $\Delta^n = \mathrm{span}_{\mathbb{Z}}\{b(x_i-1) : b \text{ in a basis of } \Delta^{n-1},\ x_i \text{ generators}\}$: no closure under further multiplication is needed, since Δ is the left ideal generated by the x_i-1 and $\Delta^{n-1}\mathbb{Z}[G] = \Delta^{n-1}$. The lattices are computed in Howell form over $\mathbb{Z}/p^E$ ($E = 30, 19, 13$ for $p = 2, 3, 5$), which is exact for our purpose: $g-1 \notin \Delta^n + p^E\Delta$ implies $g-1 \notin \Delta^n$, and the equalities $D_n = \gamma_n$ below are proved by such non-memberships (memberships are certified separately by $p^{E-1}\Delta \subseteq \Delta^n + p^E\Delta$). Groups are taken from the SmallGroups library [BEO02] of GAP [GAP]; each group is computed until $D_n = 1$.

Checks. $\gamma_n \subseteq D_n$ at every level of every group; a second method (left multiplication on conjugacy-class representatives of a pc-generating set) on all groups of orders 8–64, 81, 243, 625; an independent exact Hermite normal form over $\mathbb{Z}$ in PARI/GP [PARI] on all groups of orders 8–64, 81 and 12 groups of order 128; known values ($\Delta^n/\Delta^{n+1} \cong G$ for cyclic G, $\Delta/\Delta^2 \cong G_{ab}$). No discrepancies.

Sieves (theorems, used to skip groups): p-class $\geq$ class, so groups of p-class 2 have class ≤ 2; for 2-groups of class 3, $D_5 \subseteq D_3 = \gamma_3$, Jennings' theorem $D_n \subseteq M_n$ (the mod-2 dimension subgroups) and Passi's $D_4^2 \subseteq \gamma_4 = 1$ give $D_5 \subseteq \Omega_1(\gamma_3 \cap M_5)$, so only groups with $K = \Omega_1(\gamma_3 \cap M_5) \neq 1$ need the lattice;

for 3-groups of class 4 the analogous $K = \Omega_1(\gamma_4 \cap M_5)$, using $D_4 = \gamma_4$ for groups without 2-torsion (from Passi's $D_4^2 \subseteq \gamma_4$ [Pas68]) and Tahara's bound $D_5^6 \subseteq \gamma_5$ [Tah81, Cor. 6.11].

Proposition 5.1 (Computer-assisted)**.** *In each of the following groups $D_n(G) = \gamma_n(G)$ for all n:*

1. *all 2-groups of order $8, 16, 32, 64, 128$ (2665 groups) and all groups of order 256 of class ≥ 3 (24 328 groups);*
2. *all 3-groups of order 81, 243, 729 (586 groups);*
3. *all 1202 groups of order 3^7 of class 4 that pass the sieve, by two methods with identical results (1202 groups, 6010 levels, no discrepancies): the lattice of §5 in its C implementation and the truncated Magnus algebra of §7, item 1 (code and description: `https://gitlab.com/ispacex/cflint-core`, `docs/analysis.md`, §6.1); the numpy code of `compute/` was run on a sample of 49 of them (every 25th), with the same answer;*
4. *all 15 groups of order 625 (a control).*

Every 3-group of class 4 and order 3^5, 3^6, 3^7 (6, 71, 1309 groups) is metabelian (this is proved in Proposition 6.5, also in Lean). Moreover, $D_5(Q) = 1$ for every metabelian group Q of order 512 and class 3: the sieve leaves 2792 of the 1 587 389 such groups (out of 1 708 439 groups of order 512 and p-class ≥ 3), and for each of the 2792 the lattice computation gives $D_4(Q) = D_5(Q) = 1$.

For the metabelian groups among these, in particular all groups of items (3) and (4) (Proposition 6.5; groups of order p^4 have class ≤ 3), $D_n = \gamma_n$ for all n also follows from [Gup91]; for them the computation is a check of the code.

Order 2^{10}. This order is not in SmallGroups, and generating all 2-groups of class 3 of this order is out of reach: at rank ≥ 4 the p-group generation tree has millions of nodes already at order 2^9. Instead, the complete result at order 2^9 cuts the search down to a few thousand candidates.

Notation. $P_1(Q) = Q$, $P_{i+1}(Q) = [P_i(Q), Q]\, P_i(Q)^2$ is the lower exponent-2 central series, and the 2-class of Q is the least c with $P_{c+1}(Q) = 1$; $M_n(Q) = Q \cap (1 + \Delta_{\mathbb{F}_2}(Q)^n)$ are the mod-2 dimension subgroups, $M_n = \prod_{i2^j \geq n} \gamma_i^{2^j}$ (Jennings [Jen41], Lazard [Laz54]). If Q has 2-class $c \geq 2$, it is an *immediate descendant* of $Q/P_c(Q)$ of step size $\log_2 |P_c(Q)|$; the p-group generation algorithm [OBr90], as implemented in ANUPQ [GNO], lists all immediate descendants of a given group of a given step size, up to isomorphism.

Lemma 5.2 (minimal counterexample)**.** *Let Q be a finite 2-group of class 3 such that $D_5(Q/N) = 1$ for every central subgroup N of order 2. If $D_5(Q) \neq 1$, then*

(a) *$Z(Q)$ is cyclic and $D_5(Q) = \Omega_1(Z(Q)) = \langle n \rangle$ has order 2;*
(b) *$n \in \gamma_3(Q) \cap M_5(Q)$;*

(c) *the 2-class c of Q is at least 4, and $P_c(Q) = \langle n \rangle$;*
(d) *Q is an immediate descendant of step size 1 of $R = Q/\langle n \rangle$, a group of class 2 or 3 and 2-class $c - 1 \geq 3$ with $\gamma_3(R)$ cyclic.*

Proof. (The lemma is also formally verified, §4.)

(a) Let $N \leq Z(Q)$, $|N| = 2$. The projection $Q \to Q/N$ maps $D_5(Q)$ into $D_5(Q/N) = 1$, so $D_5(Q) \subseteq N$. Hence a nontrivial $D_5(Q)$ equals *every* central subgroup of order 2: there is only one, so $\Omega_1(Z(Q))$ has order 2, the abelian 2-group $Z(Q)$ is cyclic, and $D_5(Q) = \Omega_1(Z(Q))$. (b) $D_5 \subseteq D_3 = \gamma_3$, and $D_5(Q) \subseteq M_5(Q)$ because $\mathbb{Z}[Q] \to \mathbb{F}_2[Q]$ maps Δ^5 into $\Delta^5_{\mathbb{F}_2}$. (c) Since $\gamma_i \subseteq P_i$ and $P_i^2 \subseteq P_{i+1}$, we have $\gamma_i^{2^j} \subseteq P_{i+j}$, and $i2^j \geq 5$ implies $i+j \geq 4$. So $M_5 \subseteq P_4$, and $1 \neq n \in P_4(Q)$ gives $c \geq 4$. The subgroup $P_c(Q)$ is nontrivial, central and elementary abelian, hence equal to $\Omega_1(Z(Q)) = \langle n \rangle$. (d) By (c), $R = Q/P_c(Q)$, the step size is 1 and R has 2-class $c - 1 \geq 3$. R has class ≤ 3; if R were abelian, then $\gamma_2(Q) \subseteq \langle n \rangle \subseteq Z(Q)$ and Q would have class ≤ 2. Finally $\gamma_3(R)$ is a quotient of $\gamma_3(Q) \subseteq Z(Q)$, which is cyclic. □

Proposition 5.3 (Computer-assisted)**.** *$D_5(Q) = 1$ for every 2-group Q of order 2^{10} and class 3 (all of them are metabelian, Remark 2.2).*

Proof. Let Q be such a group with $D_5(Q) \neq 1$. For every central N of order 2 the quotient Q/N has order 2^9 and class ≤ 3, so $D_5(Q/N) = 1$: by Proposition 5.1 for class 3, and since $D_5 \subseteq D_3 = \gamma_3 = 1$ for class ≤ 2. By Lemma 5.2, Q is then isomorphic to an immediate descendant of step size 1 of a group R of order 2^9, of class 2 or 3 and 2-class ≥ 3, with $\gamma_3(R)$ cyclic, and Q has class 3, cyclic centre, and its involution n of the centre lies in $\gamma_3(Q) \cap M_5(Q)$. From here on the argument is computational. The SmallGroups library contains 1 422 862 such groups R. For each, ANUPQ (GAP 4.15.1 [GAP], ANUPQ 3.3.2 [GNO]) lists the immediate descendants of step size 1 up to isomorphism, 1 819 675 groups in all; 88 809 of them have class 3, and exactly 3400 of these have cyclic centre with involution in $\gamma_3 \cap M_5$ (by rank $d(Q) = 2, 3, 4, 5, 6$: 15, 301, 2756, 319, 9; none of rank 7). For each of the 3400 the lattice computation of §5 gives $D_4(Q) = D_5(Q) = 1$, in fact $D_n(Q) = \gamma_n(Q)$ for all n. This contradicts $D_5(Q) \neq 1$.

Certificates. The following is formally verified in Lean, with `native_decide` as explained below. Independently of the lattice computation, each of the 3400 groups (understood as the group defined by its ANUPQ pc presentation on $g_0, \ldots, g_9$) comes with a finite module $V = \bigoplus \mathbb{Z}/2^{a_i}$ and matrices for the action of the g_k (found from a truncated Magnus algebra mod 2^E; only the check matters). Lean checks that the presentation has pc shape, that the matrices are well defined and invertible on V and satisfy the pc relations, that every product $(g_{k_1} - 1) \cdots (g_{k_4} - 1)$ acts as 0 on V (checked on spanning sets: starting from the standard generators of V, the images under the $g_k - 1$ are reduced to a spanning set at each depth, and none is left at depth 4; a lemma proved in Lean and checked by the kernel shows that this suffices), and that the 2^{10} normal words act by distinct

operators. A theorem proved in Lean and checked by the kernel turns a passing check into a faithful representation whose linear extension kills Δ^4, so $D_4(Q) = D_5(Q) = 1$ and $|Q| = 2^{10}$ (`D_four_cand`, `card_cand`; directory `GroupDimension/Cert1024/`). The 3400 checks themselves run as compiled code (`native_decide`, 34 calls), not in the kernel, so they trust the Lean compiler. With GAP, the relations of every presentation were verified in the corresponding ANUPQ group, so by $|Q| = 2^{10}$ the presented group is that group. The completeness of the list of 3400 rests on Lemma 5.2(d), which is formally verified, and on the ANUPQ search, a computation that is not certified. □

Checks of the order-2^{10} search. The same pipeline at orders $2^6, \dots, 2^9$ (parents from SmallGroups of half the order) was compared with a direct pass over SmallGroups of that order: the numbers of class-3 groups with $|P_c| = 2$ and 2-class ≥ 4 (3, 27, 320, 4185), of those with cyclic centre, and of the candidates (3, 11, 64, 308) agree, also rank by rank. A second lattice method (the truncated Magnus algebra of §7, item 1, run on a pc-presentation, so independent of the multiplication table) gives the same answer on 3391 of the 3400 candidates. The search took 62.9 CPU-hours in GAP and the lattices 21.3 CPU-hours. The code and the per-chunk and per-candidate tables are in `compute/pq1024`.

Theorem 5.4 (Computer-assisted). (1) *$D_5(G) \subseteq \gamma_4(G)$ for every finite group G whose Sylow 2-subgroup has order at most 2^{10}; in particular for every group of order less than 2048.*

(2) *$D_n(G) = \gamma_n(G)$ for all n and every finite group G of order less than 512.*

(3) *$D_5(G)/\gamma_5(G)$ has no 3-torsion for every finite group G whose Sylow 3-subgroup has order at most 3^7; in fact $D_5(Q) = \gamma_5(Q)$ for every 3-group Q of order at most 3^7. (Extended to 3^9 in Theorem 6.14.)*

Proof. (1) is Remark 2.3 with Propositions 5.1 and 5.3: every metabelian 2-group of class 3 and order $\leq 2^{10}$ is covered (orders $\leq 2^8$ by the lattice for all groups, 2^9 by the sieve and the lattice, 2^{10} by Lemma 5.2, ANUPQ and the lattice). For (2) and (3): a finite nilpotent group is the direct product of its Sylow subgroups and D_n, γ_n respect direct products, and for non-nilpotent finite G one reduces to the nilpotent quotients $G/\gamma_m(G)$ (this is standard, see [MP09]); groups of order < 512 have 2-part $\leq 2^8$, 3-part $\leq 3^5$ and p-part $\leq p^3$ for $p \geq 5$, where class ≤ 2 makes everything trivial. For the 3-part in (3): if Q is a 3-group of order $\leq 3^7$, then $Q/\gamma_5(Q)$ has class ≤ 4 and order $\leq 3^7$, hence is metabelian (class ≤ 3 always is, Remark 2.2; class 4 by Proposition 5.1), and Gupta–Tahara [GT85, Thm. 3.2(c)] ($D_{p+2} = \gamma_{p+2}$ for finitely generated metabelian p-groups, p odd) gives $D_5(Q/\gamma_5) = 1$, i.e. $D_5(Q) \subseteq \gamma_5(Q)$. For the 1202 candidates of order 3^7 this gives $D_5 = \gamma_5$ by a theorem, independently of item (3) of Proposition 5.1, where $D_n = \gamma_n$ for all n was computed for each of them (for all n this also follows from [Gup91],

as they are metabelian). The reduction for non-nilpotent groups in (2)–(3) is standard, and we do not write it out. □

Observations. In every group of class c computed, $D_{c+1} = 1$ and $D_k = \gamma_k$ at each step. For metabelian groups of class 3 the Jennings sieve removes almost everything (orders 64, 128, 256, 512: 111/114, 1116/1137, 21 105/21 325, 1 584 597/1 587 389), and the survivors have $D_4 = 1$ already. For 3-groups of class 4 the sieve removes little (9/71 at order 729, 107/1309 at order 2187). Computing time: about 3.6 hours wall clock, about 30–40 CPU-hours.

Remark 5.5 (pruning the generation tree). Let Q be a finite 2-group of class 3 with cyclic centre, $d = d(Q)$, $C = C_Q(\gamma_2(Q))$, $X = \gamma_2(Q)/(\gamma_2(Q) \cap Z(Q))$. The commutator map $X \times Q/C \to \gamma_3(Q)$ is a non-degenerate bilinear pairing into the cyclic group $\gamma_3(Q) \le Z(Q)$; hence $X \cong Q/C$, a quotient of Q_{ab}, and X has rank $\le \min(d, d(d-1)/2)$. If $|\gamma_3(Q)| = 2^e$, then Q_{ab} has two invariants $\ge 2^e$ (the map $(u, v) \mapsto [u, v, z]$ factors through $\Lambda^2 Q_{ab}$) and X has an element of order $\ge 2^e$. For a minimal counterexample Q (Lemma 5.2) with central involution n this gives conditions on every quotient $V = Q/N$ with $N \le \Phi(Q)$ (with $n \in N \le \Phi(Q)$ for the second), which pass to further quotients and so may be used to cut whole subtrees of the p-group generation tree: (i) there is a cyclic K, $\gamma_3(V) \le K \le \gamma_2(V) \cap Z(V)$, with $\mathrm{rk}(\gamma_2(V)/K) \le \min(d, d(d-1)/2)$; (ii) for such a K, $|Q| \ge 2|K|\,|A \vee B \vee E_1|\,|A \vee V_{ab} \vee E_2|$, where $A = \gamma_2(V)/K$, $B = V/C_V(\gamma_2(V))$, $E_1 = \mathbb{Z}/2^{e_V+1}$, $E_2 = E_1^2$, $2^{e_V} = |\gamma_3(V)|$, and $\vee$ is the componentwise maximum of abelian invariants. In particular every such V of class 3 has $d + 4e_V + 2 \le \log_2 |Q|$. These conditions are proved by hand; the proofs are in `compute/pq2048/README.md`. Condition (i) is vacuous for $d \le 3$ and below order 2^{2d+1}; condition (ii) is the one that cuts.

Order 2^{11}*, ranks* 2–7. At order 2^{11} the same argument needs as parents the groups of order 2^{10} of class ≤ 3 with cyclic γ_3; they are not in SmallGroups, and their number is estimated at 0.5–$1.1 \cdot 10^9$. Remark 5.5 cuts whole subtrees without losing a candidate (checked: the same 3, 11, 64, 308 candidates at orders 2^6–2^9) and leaves, by a sample, 10^5–$2 \cdot 10^6$ parents; without rules (A), (B), (T) below, generating them would take an estimated 100 CPU-days, mostly at rank 5. Ranks 2–7 have been run through to order 2^{11}: 17, 586, 52 568, 824, 13 and 9 candidates (at ranks 5 and 6 after rule (T) below), all with $D_4 = D_5 = 1$ by computation (with the second lattice method described after Proposition 5.3). For rank 4 the walk passes through 2 756 candidates at order 2^{10}, exactly the rank-4 count of Proposition 5.3; it kept 52 181 parents of order 2^{10} and cut 9 937, 136 127, 946 663 vertices at orders 2^8, 2^9, 2^{10} (code: `compute/pq2048`). The pruned search used there was checked at order 2^{10}, where it reproduces the 3400 candidates of Proposition 5.3 exactly, rank by rank, while keeping 4.2% of the parents. Ranks 5 and 6 used, on top of Remark 5.5, three more rules, each proved by hand;

the proofs and references are in `compute/pq2048/README.md`. (A) A vertex of order 2^9, 2-class 2 and elementary abelianization is not expanded: it is not a parent, its steps ≥ 2 go past order 2^{10}, and its step-1 descendants all have class 3 and fail the bound of Remark 5.5(ii). (B) The step-1 descendants of class 3 of a vertex of order 2^9 and 2-class 2 whose abelianization has fewer than two invariants ≥ 4 are dropped, as they fail the same bound. (T) A leaf of order 2^{10} whose abelianization has at most one invariant ≥ 4 is dropped: a candidate Q above it has the same abelianization, so $D_4(Q) = \gamma_4(Q) = 1$ by Tahara's Corollary 19 [Tah77a, p. 125] (for a 2-group of class 3 whose abelianization has at most one cyclic factor of order ≥ 4, $D_4 = \gamma_4$; Tahara attributes it to Passi [Pas79, Thm. 10.1]) and hence $D_5(Q) = 1$. Rules (A) and (B) lose no candidate. Rule (T) does drop candidates, but only ones with $D_5 = 1$, so the rank-5 and rank-6 counts above are of the candidates left after it; a rank-6 run that applied (T) to only part of its pieces kept 469 candidates, all with $D_4 = D_5 = 1$ by computation. These two ranks were run with ANUPQ with the fixes of the audit mentioned in the proof of Theorem 6.14, and with automorphism groups not computed for descendants that are leaves (an option proposed upstream as gap-packages/anupq#124; on 283 test groups this version gives the same 35 213 descendants as without the option). Ranks 2–4 and 7, like the order-2^{10} search of Proposition 5.3, were first run with ANUPQ before those fixes (two of the fixed defects can change the descendants) and then repeated with the same fixed build: the results agree exactly, namely the results of all 858 pieces of the search and the 3400 candidates with their lattice answers at order 2^{10}, and the candidates of ranks 2–4 and 7 at order 2^{11}; so does the direct generation at rank 8 described after Proposition 5.7. For rank 5 the walk passes through 319 candidates at order 2^{10}, and for rank 6 through 9, again exactly the counts of Proposition 5.3; rank 5 kept 206 519 parents of order 2^{10}. The tree searches took about 15 CPU-hours at rank 4 and 83 at rank 5. At rank 7 the unmodified ANUPQ fails with an integer overflow on the step-2 descendants of $(\mathbb{Z}/2)^7$; these are the 1 112 groups of order 2^9, rank 7 and 2-class 2, a contiguous block of SmallGroups (by `SmallGroupsInformation`), and the walk starts from them. Ranks ≥ 8 need no search (Proposition 5.7 below); so order 2^{11} is settled (Proposition 5.10).

Order 2^{11}, rank ≥ 8. Call a 2-group Q a *candidate* if it has class 3, cyclic centre with involution $n \in \gamma_3(Q) \cap M_5(Q)$, and 2-class $c \geq 4$ with $P_c(Q) = \langle n \rangle$. By Lemma 5.2 and Proposition 5.3, a 2-group of order 2^{11} and class 3 with $D_5 \neq 1$ is a candidate (its quotients by central subgroups of order 2 have order 2^{10} and class ≤ 3).

Lemma 5.6. *There is no candidate Q of order 2^{11} with $Q/P_3(Q) \cong \mathbb{Z}/4 \times (\mathbb{Z}/2)^7$.*

Proof. (The lemma is also formally verified, §4.)

Such a Q has rank 8, so $|P_2(Q)| = |\Phi(Q)| = 8$; the series $P_2 \supsetneq P_3 \supsetneq P_4 \supseteq P_c \neq 1$ decreases strictly until it reaches 1, hence $|P_3(Q)| = 4$ and

$P_4(Q) = \langle n \rangle$. (1) Q/P_3 is abelian, so $\gamma_2(Q) \le P_3(Q)$. (2) Q has class 3, so $\gamma_3 = [\gamma_2, Q] \ne 1$, and $\gamma_3 \ne \gamma_2$ since Q is nilpotent and $\gamma_2 \ne 1$. Hence $|\gamma_2| = 4$, $\gamma_2 = P_3$, $|\gamma_3| = 2$, and $\gamma_3 = \langle n \rangle$, the only central subgroup of order 2. (3) $M_5(Q) = \prod_{i2^j \ge 5} \gamma_i^{2^j} = Q^8 \gamma_2^4 \gamma_3^2 = Q^8$, since $\gamma_4 = 1$, $|\gamma_2| = 4$ and $|\gamma_3| = 2$. (4) Q/P_3 has exponent 4, so $Q^4 \le P_3$ and $Q^8 \le P_3^2 \le P_4 = \langle n \rangle$. (5) As $n \in M_5(Q) = Q^8$, there is $x \in Q$ with $x^8 = n$. Then $x^4 \in P_3 = \gamma_2$ has order 4, so $\gamma_2 = \langle x^4 \rangle$. (6) For $g \in Q$, $[x, g] \in \gamma_2$, γ_2 is abelian and γ_3 is central, so $[x, g]^{x^i} = [x, g]\,[x, g, x]^i$ and

$$[x^k, g] = \prod_{i<k} [x, g]^{x^i} = [x, g]^k\,[x, g, x]^{k(k-1)/2}.$$

For $k = 4$ this is $[x, g]^4 [x, g, x]^6 = 1$, because γ_2 has exponent dividing 4 and $|\gamma_3| = 2$. So x^4 is central. (7) Then $\gamma_2 = \langle x^4 \rangle \le Z(Q)$ and $\gamma_3 = [\gamma_2, Q] = 1$, contradicting class 3. □

Proposition 5.7. *$D_5(Q) = 1$ for every 2-group Q of order 2^{11}, class 3 and rank $d(Q) \ge 8$. More generally, every candidate Q has $|\Phi(Q)| \ge 16$, that is, $d(Q) \le \log_2 |Q| - 4$.*

Proof. (The second statement is proved by hand and is also formally verified, and so is the deduction of the first statement from the second and Proposition 5.3, see §4; through Proposition 5.3 the first statement is computer-assisted.)

The first statement follows from the second: a counterexample would be a candidate (see above) with $|\Phi(Q)| = 2^{11-d} \le 8$. Let Q be a candidate with $|\Phi(Q)| \le 8$. The series P_i decreases strictly until it reaches 1 (if $P_{i+1} = P_i$, it is constant from i on), and $\Phi(Q) = P_2$ (Burnside); since $c \ge 4$, $P_2 \supsetneq P_3 \supsetneq P_4 \supseteq P_c = \langle n \rangle$ gives $|P_2| = 8$, $|P_3| = 4$ and $P_4 = \langle n \rangle$. Also $\gamma_2 \le P_2$, $\gamma_3 \le P_3$, and γ_3 is central since $\gamma_4 = 1$; so $[u^m, g] = [u, g]^m$ for $u \in \gamma_2$, $g \in Q$, and γ_3 is generated by the $[u, g]$. As in step (2) of the proof of Lemma 5.6, $\gamma_3 \ne 1$ and $\gamma_3 \ne \gamma_2$, so $|\gamma_2| \ge 4$. Since $\gamma_4 = 1$, $M_5(Q) = Q^8 \gamma_2^4 \gamma_3^2$, and the generators x^8, y^4 ($y \in \gamma_2 \le P_2$) and z^2 ($z \in \gamma_3 \le P_3$) of the three factors lie in $P_4 = \{1, n\}$. As $n \in M_5(Q)$, one of them equals n.

(a) $z^2 = n$ for some $z \in \gamma_3$. Then z has order 4, so $\gamma_3 = P_3 = \langle z \rangle$ is central. For $u \in \gamma_2$ and $g \in Q$ we have $u^2 \in P_3$, hence $[u, g]^2 = [u^2, g] = 1$ and $[u, g] \in \langle z^2 \rangle = \langle n \rangle$. So $\gamma_3 \le \langle n \rangle$, contradicting $z^2 = n \ne 1$.

(b) $y^4 = n$ for some $y \in \gamma_2$. Then y has order 8, so $\gamma_2 = P_2 = \langle y \rangle$ and $P_3 = \langle y^2 \rangle$. For $g \in Q$, $[y, g]^2 = [y^2, g] \in [P_3, Q] \le P_4 = \langle n \rangle$. If $[y, g]^2 = n$, then $[y, g] \in \gamma_3$ has order 4, so $\langle [y, g] \rangle = P_3 \ni y^2$, and $y^2 \in \gamma_3$ is central: $[y^2, g] = 1$, a contradiction. Hence $[y, g]^2 = 1$, that is, $[y, g] \in \langle y^4 \rangle = \langle n \rangle$, and γ_3, generated by the $[y^m, g] = [y, g]^m$, lies in $\langle n \rangle$. Now let $g, h \in Q$. Since $g^2 \in P_2 = \gamma_2$, we have $[g^2, h] \in \gamma_3$, and $[g^2, h] \equiv [g, h]^2 \pmod{\gamma_3}$; so $[g, h]^2 \in \gamma_3 \le \langle n \rangle$, $[g, h]^4 = 1$ and $[g, h] \in \langle y^2 \rangle$. Then $\gamma_2 \le \langle y^2 \rangle$ has order ≤ 4, a contradiction.

(c) $x^8 = n$ for some $x \in Q$. Then $x^2 \in P_2$ has order 8, so $P_2 = \langle x^2 \rangle$ is cyclic, and $\gamma_2 \le P_2$ has order 4 or 8. If $|\gamma_2| = 8$, then $\gamma_2 = P_2 \ni x^2$ and

$(x^2)^4 = n$, which is case (b). If $|\gamma_2| = 4$, then $\gamma_2 = \langle x^4 \rangle = P_3$, and steps (2)–(7) of the proof of Lemma 5.6 give a contradiction: they use $Q/P_3(Q)$ only through $\gamma_2 \le P_3$, $|P_3| = 4$, $P_4 = \langle n \rangle$ and $Q^4 \le P_3$, which holds in every group ($x^4 = (x^2)^2$ with $x^2 \in P_2$). □

The bound is sharp at both searched orders: the candidates of order 2^{10} have rank ≤ 6, with 9 of rank 6 (Proposition 5.3), and at order 2^{11} there are 9 of rank 7, while direct generation at rank 8 finds none (below).

Checks. The case $d = 8$ (then $c = 4$, $|P_3(Q)| = 4$ and $P_4(Q) = \langle n \rangle$, as in the first paragraph of the proof) was first settled through $V = Q/P_3(Q)$, of order 2^9, rank 8 and 2-class 2: $R = Q/\langle n \rangle$ is an immediate descendant of step 1 of V, of class 2 by Remark 5.5(ii) (if it had class 3, then $d + 4e_R + 2 \le 11$ with $e_R \ge 1$, i.e. $d \le 5$). In the SmallGroups library the groups of order 2^9 and rank 8 are the 12 groups SmallGroup$(512, i)$, $10\,494\,201 \le i \le 10\,494\,212$ (a group V of rank n with $|\Phi(V)| = 2$ is determined by its squaring map $q\colon V/\Phi(V) \to \Phi(V)$, a nonzero quadratic form on $\mathbb{F}_2^n$ whose polar form is the commutator form, so such groups correspond to nonzero quadratic forms up to equivalence; if the polar form has rank $2k$ and radical R, then $q|_R = 0$ gives two classes for each $1 \le k \le n/2$ (Arf invariant 0 or 1) and $q|_R \ne 0$ one class for each $0 \le k < n/2$, so $2\lfloor n/2 \rfloor + \lceil n/2 \rceil$ classes, 12 for $n = 8$; the same formula gives the numbers of groups of order 2^{n+1} and rank n in SmallGroups for $n = 2, \dots, 7$, namely $3, 4, 6, 7, 9, 10$). Of the 11 non-abelian ones, 10 have trivial nucleus and hence no immediate descendants at all [OBr90]; for SmallGroup$(512, 10\,494\,202)$ the nucleus has dimension 1, but every immediate descendant of class ≤ 3 has class 3, so it has no descendant R of class 2. Both are read off the p-covering group, without computing automorphism groups; the two rules, with proofs by hand, are in `compute/pq2048/README.md` (Rules 0 and 2), where they are also checked against full ANUPQ expansion. The remaining group SmallGroup$(512, 10\,494\,201) \cong \mathbb{Z}/4 \times (\mathbb{Z}/2)^7$ is excluded by Lemma 5.6.

An independent check of rank 8 by direct generation agrees: on $\mathbb{Z}/4 \times (\mathbb{Z}/2)^7$ the unmodified ANUPQ needs an estimated 20 GB of memory, while a version with reduced memory use completes it in 2.1 GB, finding 8 immediate descendants of order 2^{10}; the walk from all 12 groups then gives no candidate of order 2^{11}.

Central products. A search beyond order 2^{11} may skip candidates that split as central products with a small factor. Let E be a finite 2-group of class 2 with $E/Z(E) \cong (\mathbb{Z}/2)^2$ and cyclic centre. Replacing E by the subgroup generated by two lifts a, b of a basis of $E/Z(E)$, which does not change the central product below, one has $E \in \{D_8, Q_8\}$ or $E \cong M_{2^{k+2}} = \langle a, b \mid a^{2^{k+1}} = b^2 = 1,\ a^b = a^{1+2^k} \rangle$, $k \ge 2$, with $Z(E) = \langle a^2 \rangle$ (an elementary verification, which we omit). For an embedding $\varphi\colon Z(E) \to Z(H)$ let $H \circ E = (H \times E)/\{(\varphi(z), z^{-1})\}$, $t = \varphi([a, b])$ and, for $E = M_{2^{k+2}}$, $s = \varphi(a^2)$.

Proposition 5.8. *Let H be a group with $t \in \gamma_2(H)$, and $n \ge 2$.*

(1) *If $E \in \{D_8, Q_8\}$, then $D_n(H \circ E) = D_n(H)$.*

(2) *If $E = M_{2^{k+2}}$ and for some $g \in H$ the element $c = g^{-2}s$ satisfies $c^{2^j} - 1 \in \Delta(H)^{j+2}$ for all $j \geq 0$ (for instance $c^{2^j} \in \gamma_{j+2}(H)$), then $D_n(H \circ E) = D_n(H)$. This holds in particular if $s = g^2$ for some $g \in H$, central or not ($c = 1$), and if $s \in \gamma_2(H)$, $s^2 \in \gamma_3(H)$ ($g = 1$, $k = 2$).*

Proof. (The proposition is also formally verified, in a slightly different form, see §4.)

Let $Q \geq H$ be one of the groups below, generated by H and elements commuting with H, and $\gamma_2(Q) = \gamma_2(H)$. Then $D_n(Q) \subseteq \gamma_2(Q) \subseteq H$ and $D_n(H) \subseteq D_n(Q)$, so $D_n(Q) = D_n(H)$ follows once $\mathbb{Z}[H] \cap \Delta(Q)^n \subseteq \Delta(H)^n$. For that it suffices to have a left $\mathbb{Z}[H]$-linear $q\colon \mathbb{Z}[Q] \to \mathbb{Z}[H]$, the identity on $\mathbb{Z}[H]$, with $q(\Delta(Q)^n) \subseteq \Delta(H)^n$. Here $\gamma_2(H \circ E) = \gamma_2(H)\langle t\rangle = \gamma_2(H)$.

(1) Let $E = D_8$, $c = [a, b]$, $\tau = c - 1$ (so $\tau^2 = -2\tau$), and $N = \mathbb{Z}[E/\langle b\rangle]$, free over $\mathbb{Z}[\langle c\rangle]$ on e_0 and $f = ae_0 - e_0$. Then $\mathbb{Z}[H] \otimes_{\mathbb{Z}[\langle c\rangle]} N$ is a $\mathbb{Z}[Q]$-module, and $q(u) = (\mathrm{id} \otimes p)(u(1 \otimes e_0))$ with $p(xe_0 + yf) = x$ is as required. Indeed, $\Delta(Q)^n$ is spanned by products of i factors from $\Delta(H)$ and $n - i$ from $\Delta(E)$, and on N in the coordinates (x, y) one has $a - 1\colon (x, y) \mapsto (0, x - 2y)$ and $b - 1\colon (x, y) \mapsto (\tau y, \tau y)$. By induction, $p(\Delta(E)^j N) = \tau^{\lceil j/2\rceil}\mathbb{Z}[\langle c\rangle]$ for $j \geq 1$; as $t - 1 \in \Delta(H)^2$, $q(\Delta(Q)^n) \subseteq \sum_i \Delta(H)^i (t-1)^{\lceil (n-i)/2\rceil} \subseteq \Delta(H)^n$. For Q_8: $u = ab \in D_8$ has $u^2 = t$, so $H' = H\langle u\rangle \leq H \circ D_8$ and $D_n(H') = D_n(H)$. If a', b' are the generators of the factor D_8 of $H' \circ D_8$, the elements $a'u, b'u$ generate a Q_8 meeting H' in $\langle t\rangle$, so $H \circ Q_8 \leq H' \circ D_8$ and $D_n(H \circ Q_8) \subseteq D_n(H') \cap H = D_n(H)$, using (1) for H'.

(2) Let $H'' = H\langle r\rangle$ with r central, $r^2 = s$, $\langle r\rangle \cap H = \langle s\rangle$. With a', b' the generators of the factor D_8 of $H'' \circ D_8$, the elements $a'r, b'$ generate a copy of E meeting H'' in $\langle s\rangle$, and $H\langle a\rangle \cong H''$ inside $H \circ E$; by (1), $D_n(H \circ E) = D_n(H'')$. Now $\mathbb{Z}[H''] = \mathbb{Z}[H] \oplus \mathbb{Z}[H]r$; put $q(u_0 + u_1 r) = u_0 + u_1 g$. As r is central, $\Delta(H'')^n = \sum_i \Delta(H)^{n-i}(r-1)^i \mathbb{Z}[H'']$, so it suffices that $f_i = q((r-1)^i) \in \Delta(H)^i$ for all i. In the abelian group $\langle g, s, r\rangle$ put $y = rg^{-1}$, so $y^2 = c$, and $x = g - 1$, $\eta = y - 1$, $\tau = c - 1$. Then $r - 1 = x + \eta + x\eta$ and $f_i = \sum_{\alpha+\beta \geq i} m_{\alpha\beta}\, x^\alpha e_\beta$ with $e_b = \pi(\eta^b)$, $\pi(v_0 + v_1 y) = v_0 + v_1$. From $\eta^2 = \tau - 2\eta$ we get $e_1 = 0$, $e_2 = \tau$ and $e_{b+2} = \tau e_b - 2e_{b+1}$. The identity $2(c^{2^j} - 1) = (c^{2^{j+1}} - 1) - (c^{2^j} - 1)^2$ and the hypothesis give $2^m \tau \in \Delta(H)^{m+2}$ for all m. Hence, by induction on $b \geq 1$, $2^m e_b \in \Delta(H)^{b+m}$ for all m; in particular $e_b \in \Delta(H)^b$. □

The condition $c \in \gamma_2(H)$, i.e. $s \in \Phi(H)$ for a 2-group, cannot be dropped from the method: if $s \notin \Phi(H)$, no such q for H'' satisfies $q(\Delta(H'')^2) \subseteq \Delta(H)^2$. Indeed, $(r-1)^2 = (s-1) - 2(r-1)$ would give $(s-1) - 2x \in \Delta(H)^2$ with $x = q(r) - 1$; applying the augmentation shows $x \in \Delta(H)$, so $s - 1 \in 2\Delta(H) + \Delta(H)^2$, i.e. $s \in H^2\gamma_2(H) = \Phi(H)$. A computation shows that for the 386 candidates H of orders 2^6–2^9, $t = n \in \gamma_3(H)$. The 291 with $|Z(H)| \geq 8$ have s a square in the cyclic $Z(H)$. Of the 95 with $Z(H) \cong \mathbb{Z}/4$, 78 have $s \in \gamma_2(H)$ and $s^2 \in \gamma_3(H)$, and the other 17 have $s = g^2$ with $g \in H$

not central. So by Proposition 5.8, $D_n(H \circ E) = D_n(H)$ for all of them, all n and $E \in \{D_8, Q_8, M_{16}\}$. Independently, the lattice computation gives $D_4 = D_5 = 1$ for all 1 544 such products.

Gluing along a proper subgroup. For $C \le Z(E)$ and an embedding $\varphi\colon C \to Z(H)$ let $H \circ_C E = (H \times E)/\{(\varphi(z), z^{-1})\colon z \in C\}$. Then $H' = HZ(E)$ centralises E, $H \circ_C E = H' \circ E$ is a gluing along all of $Z(E)$ (with φ the inclusion, so $s = a^2 \in H'$ for $E = M_{2^{k+2}}$), and $\gamma_i(H') = \gamma_i(H)$ for $i \ge 2$. For $E = M_{2^{k+2}}$, $H' \circ E$ can be written as $H \circ_C E$ with $C \ne Z(E)$ exactly when $s \notin \Phi(H')$, which is where the method of part (2) for H' fails by the remark above. The projection then has to go to H instead of H'.

Lemma 5.9. *Let $K = H\langle r\rangle$ with r central, $|K : H| = m$, and $r^m = \sigma = g^m$ for some $g \in H$, not necessarily central. Then $\mathbb{Z}[H] \cap \Delta(K)^n = \Delta(H)^n$ for all n, and $D_n(K) = D_n(H)$ for $n \ge 2$.*

Proof. (The lemma is also formally verified, §4.)

$\mathbb{Z}[K] = \bigoplus_{0 \le j < m} \mathbb{Z}[H] r^j$; put $q(\sum_j u_j r^j) = \sum_j u_j g^j$. As $r^m = g^m \in H$, $q(ur^j) = ug^j$ for all $j \in \mathbb{Z}$, so $q(u\,p(r)) = u\,p(g)$ for every Laurent polynomial p. As r is central, $\Delta(K)^n$ is spanned by the $x(r-1)^i r^j$ with $x \in \Delta(H)^{n-i}$, and $q(x(r-1)^i r^j) = x(g-1)^i g^j \in \Delta(H)^n$. Hence $\mathbb{Z}[H] \cap \Delta(K)^n = q(\mathbb{Z}[H] \cap \Delta(K)^n) \subseteq \Delta(H)^n$; the reverse inclusion is clear, and $D_n(K) \subseteq \gamma_2(K) = \gamma_2(H) \subseteq H$ gives the rest. □

For $m = 2$ this is the case $c = 1$ of Proposition 5.8(2). Let $E = M_{2^{k+2}}$, $C = \langle a^{2^{j+1}}\rangle$ with $1 \le j < k$, and $\sigma = \varphi(a^{2^{j+1}})$. If $t \in \gamma_2(H)$ and $\sigma = g^{2^{j+1}}$ for some $g \in H$, then $D_n(H \circ_C E) = D_n(H)$ for $n \ge 2$: in the proof of (2) for H' one has $H'' = H'\langle r\rangle = H\langle r\rangle$ with $r^{2^{j+1}} = \sigma$ and $\langle r\rangle \cap H = \langle\sigma\rangle$, so $|H'' : H| = 2^{j+1}$, and Lemma 5.9 applies with $m = 2^{j+1}$. For instance, $H \circ_{\langle t\rangle} M_{16}$ needs $t = g^4$. The hypothesis $t \in \gamma_2(H)$ costs nothing for the search: if $H \circ_C E$ is a candidate, t is the involution of its cyclic centre, so $t = n \in \gamma_3(H \circ_C E) = \gamma_3(H)$. A computation shows that, of the 386 candidates of orders 2^6–2^9, 20 split as central products (5 of order 2^8, 15 of order 2^9), with 212 factors E in all. The gluing is proper in 16 of these splittings, belonging to 4 candidates, always with M_{32} or M_{64}; Lemma 5.9 covers 4 of them. Each of the 20 candidates also has a splitting covered by Proposition 5.8, so at these orders a search that skips split candidates does not need proper gluing (code: `compute/central`). The other 12 splittings would need an analogue of (2) for $m = 4$ (taking H maximal in H', i.e. $j = 1$) with $g^{-4}\sigma \ne 1$; it cannot go through a projection to H in general, since for $Q = D_8 \circ_{\langle t\rangle} M_{16}$ one has $\mathbb{Z}[D_8] \cap \Delta(Q)^n \ne \Delta(D_8)^n$ at $n = 4, 5$ (index 4), although $D_n(Q) = D_n(D_8)$ by computation. Whether such an analogue holds is open.

3-torsion. For 3-torsion the computation is no longer needed up to order 3^7 (Theorem 5.4(3)); orders 3^8 and 3^9 and an infinite family are treated in §6.

Sections of Rips's group. One may hope to gauge how large a counterexample to Question 1.1 must be from the smallest sections of the known examples with $D_4 \neq \gamma_4$. For Rips's group we found no such section of order below 2^{38}: proper subgroups are settled, quotients only sampled, and proper sections H/N with $H < G$, $N \neq 1$ are not checked. We take the group from Tahara's presentation, the member $k = 2$, $\ell = 0$ of his family of order $2^{8k+22+\ell}$ [Tah78, §3], with generators $x_1, \dots, x_4$, $y_1, \dots, y_4$. It has order 2^{38} and class 3, with $D_4 = \{1, w\}$, $w = y_1^{128}$, as Tahara states, and $\Omega_1(Z) = \langle w \rangle$. All 15 maximal subgroups and $G/\langle w \rangle$ have $D_4 = 1$ (computed in the truncated Magnus algebra on all of γ_3, which contains D_4; an answer $D_4 = 1$ of this computation is exact, and $D_4 \neq 1$ is certified). Hence every proper subgroup has $D_4 = 1$ (as $D_4(H) \subseteq D_4(M)$ for $H \leq M$), and every proper quotient is a quotient of $G/\langle w \rangle$. A random sample of 26 147 quotients, not necessarily distinct, taken along descending chains, gave no new $D_4 \neq 1$. A heuristic search over the exponents of Tahara's presentation found no group of that shape of order below 2^{38} with $D_4 \neq 1$. That every proper quotient has $D_4 = 1$ is checked on a sample only (code: `compute/magnus`).

A lower bound for $D_4 \neq \gamma_4$. Rips's group, of order 2^{38}, is the smallest example of $D_4 \neq \gamma_4$ we know; below it we can at least say where an example cannot be: *a 2-group with $D_4 \neq \gamma_4$ has order $\geq 2^{11}$, so $D_4(G) = \gamma_4(G)$ for every finite group G whose Sylow 2-subgroup has order $\leq 2^{10}$* (computer-assisted). The second statement follows from the first as in Remark 2.3: $D_4(G/\gamma_4(G))$ has exponent 2 by Passi's $D_4^2 \subseteq \gamma_4$ [Pas68], so it lies in the Sylow 2-subgroup S, a direct factor of the nilpotent group $G/\gamma_4(G)$, and the projection onto S maps it into $D_4(S) = \gamma_4(S) = 1$. For the first, let Q be a 2-group of least order with $D_4 \neq \gamma_4$; by Proposition 5.1(1) its order is $\geq 2^9$ (for class ≤ 2, $D_4 \subseteq D_3 = \gamma_3 = 1$). As in Lemma 5.2, Q has class 3 (Q/γ_4 is again an example), every proper quotient has $D_4 = 1$, so the centre is cyclic and $D_4(Q) = \Omega_1(Z(Q)) = \langle n \rangle$ with $n \in \gamma_3(Q) \cap M_4(Q)$. Since $M_4 = Q^4\gamma_2^2\gamma_4 \leq P_3$, the 2-class c of Q is ≥ 3, $P_c(Q) = \langle n \rangle$, and Q is a step-1 immediate descendant of $R = Q/\langle n \rangle$, of class 2 or 3, 2-class ≥ 2 and with $\gamma_3(R)$ cyclic. At order 2^9, 4 421 groups in SmallGroups satisfy these conditions, and all have $n \notin D_4$ (truncated Magnus algebra; exact). At order 2^{10} the parents R are the 10 208 630 groups of order 2^9 with these properties, and their step-1 descendants are R^*/U, where R^* is the p-covering group, $L \leq Z(R^*)$ its multiplier, and U runs over the hyperplanes of L not containing the nucleus. The conditions class 3, $n \in \gamma_3$ and $n \in M_4$ are linear in U (of the form $X \not\leq U$ for a subgroup X of L), and so is $n \in D_4$: with $A = \mathbb{Z}[R^*]/\Delta^4$ and $V = \sum_{m \in L}(m-1)A$, the map $\psi\colon L \to V/V\Delta$, $m \mapsto m - 1$, is a homomorphism, since L is central and $uv - 1 = (u-1) + (v-1) + (u-1)(v-1)$; as $\mathbb{Z}[R^*/U]/\Delta^4 = A/\sum_{u \in U}(u-1)A$, Nakayama's lemma gives $n \in D_4(R^*/U)$ if and only if $\ker\psi \not\leq U$. For 1 253 329 parents some U satisfies these linear conditions (cyclicity of the centre is not imposed, which only enlarges the search), and for none of them does a U with $n \in D_4$ exist. We compute $\ker\psi$ modulo 2^{30}, which can only enlarge it, so this negative answer is exact. As

a control, Rips's group in Tahara's presentation, as a step-1 descendant of $G/\langle w\rangle$, is found with $\dim\ker\psi = 1$; and at orders 2^8, 2^9 the method agrees with the direct sieve `d4s9.g` (code: `compute/pq1024`, `run_d4lin.sh`, and `compute/magnus`, `d4s9.g`; about 25 hours on 8 cores).

Sections of the Bartholdi–Mikhailov 3-group. This concerns 3-torsion (§7, item 2) rather than Question 1.1. The 3-group G of order 3^{494} of [BM23, §9.3] is an example of $D_7 \neq \gamma_7$, not of $D_4 \neq \gamma_4$. By computation, it has class 6, so $\gamma_7 = 1$, and its element ω has order 3. A computation shows that the relations of the group (9.2) of [BM23] hold in G for BM's choices of y_i, y_{ij}, so (9.2) maps to G (not onto) with ω the image of its ω; hence $\omega \in D_7(G)$ by [BM23, Prop. 9.2], as D_7 is functorial. The centre of G is computed to be $C_3 \times C_{3^{15}}$, with ω in its Frattini subgroup; so every maximal subgroup of the centre contains ω, and quotienting by such subgroups, as in [BM23], cannot go further. A central subgroup $\langle u\rangle$ of order 3 with $u \notin \langle\omega\rangle$ can still be factored out: $G/\langle u\rangle$ has order 3^{493}, $\omega \neq 1$ in it, hence $D_7 \neq \gamma_7$, and its centre is cyclic, so every proper quotient of it kills ω (these facts are computed; code: `compute/magnus/bm494`). The proof of [BM23, Prop. 9.2] uses only the relations (9.2), which are identities in G and so hold in any subgroup, and that $e_0, \dots, e_3$ (here $x_1, \dots, x_4$) lie in the group, that $e_i^{3^{2i}} \in \gamma_2$, and that the elements y_{ij} it uses lie in γ_3; by computation, in G and in each $G/\langle u\rangle$ every maximal subgroup M containing $x_1, \dots, x_4$ violates $e_i^{3^{2i}} \in \gamma_2(M)$ for some i, so this argument gives $\omega \in D_7(H)$ for no proper subgroup $H \ni x_1, \dots, x_4$ of G or of $G/\langle u\rangle$. Nor does this argument apply to any section M/N of such an M in which ω survives: by computation, if $e_i^{3^{2i}} \in \gamma_2(M)N$ for all i (the second condition in M/N), then $\omega \in N$. Indeed, by computation $A = \gamma_6$ of G and of each $G/\langle u\rangle$ is central and cyclic of order 3^6 with socle $\langle\omega\rangle$, so $\omega \notin N$ forces $N \cap A = 1$; the condition gives a series $\Gamma_j \leq \gamma_j(M)N$ starting from M and $\langle x_1, x_2^9, x_3^{81}, x_4^{729}\rangle\gamma_2(M)$, and the homomorphisms $g \mapsto [g, m_1, \dots, m_{6-j}]$ from γ_j of G (resp. $G/\langle u\rangle$) to A ($m_k \in M$) then turn $\Gamma_j \leq \gamma_j(M)N$ into linear conditions over $\mathbb{Z}/3^6$, which fail in all $4 \cdot 40 = 160$ cases. The same computation with H in place of M gives this for every H with $\Phi(Q)\langle x_1, \dots, x_4\rangle \leq H < Q$ of index ≥ 9, where Q is G or one of the $G/\langle u\rangle$ (171 such H in each Q, 684 in all): if $e_i^{3^{2i}} \in \gamma_2(H)N$ for all i, then $\omega \in N$. Sections H/N with $H \not\supseteq \Phi(Q)$, other arguments, and elements of D_7 of quotients of G that are not images of ω are not considered here.

Order 2^{11}, all ranks.

Proposition 5.10 (Computer-assisted)**.** *$D_5(Q) = 1$ for every 2-group Q of order 2^{11} and class 3. Hence a counterexample to Question 1.1 among finite 2-groups of class 3 has order at least 2^{12}.*

Proof. A 2-group of order 2^{11} and class 3 with $D_5 \neq 1$ is a candidate (above). Ranks ≥ 8: Proposition 5.7. Ranks 2–7: the searches above find every candidate of these ranks except, at ranks 5 and 6, those dropped by rule (T), which have $D_5 = 1$ by [Tah77a, Cor. 19]; all the candidates found have $D_5 =$

1. Rank 1: cyclic groups are abelian. Orders below 2^{11}: Propositions 5.1 and 5.3. See also the remark on the ANUPQ versions above. □

Since order 2^{11} is settled, Lemma 5.2 applies to every 2-group Q of order 2^{12} and class 3 with $D_5(Q) \neq 1$ (its quotients by central subgroups of order 2 have order 2^{11} and class ≤ 3). So such a Q is a candidate, and by Proposition 5.7 it has rank $d(Q) \leq 8$; this, too, is computer-assisted.

6. 3-torsion: orders 3^8 and 3^9, and groups with G/γ_2 of exponent 3

Theorem 5.4(3) settles 3-torsion up to order 3^7 by metabelianity. Here we go further: we prove $D_5 = \gamma_5$ for all 3-groups of order $\leq 3^9$ (Theorem 6.14) and, without computer assistance, for every finite 3-group Q with $Q/\gamma_2(Q)$ of exponent 3, of any order (Theorem 6.6). Code and data are in `compute/magnus`. Throughout, the *type* of a group Q of class c is $(a_1, \dots, a_c)$ with $p^{a_i} = |\gamma_i(Q)/\gamma_{i+1}(Q)|$, and $L(Q) = \bigoplus_i \gamma_i/\gamma_{i+1}$ is its Lie ring.

Lemma 6.1. *Let p be odd and Q a finite p-group of class $\leq p+1$. Then $D_{p+2}(Q) \subseteq Q'' \cap M_{p+2}(Q)$. In particular, for $p = 3$ and class ≤ 4, $D_5(Q) \subseteq Q'' \cap M_5(Q)$, and $D_5(Q) \neq 1$ needs $Q'' \neq 1$, hence $d(Q) \geq 3$.*

Proof. (The lemma is also formally verified, with [GT85, Thm. 3.2(c)] as a hypothesis, §4.)

Q/Q'' is metabelian of class $\leq p+1$, so $D_{p+2}(Q/Q'') = \gamma_{p+2}(Q/Q'') = 1$ by [GT85, Thm. 3.2(c)] (indeed $D_n(Q/Q'') = \gamma_n(Q/Q'')$ for all n by [Gup91]), and $D_{p+2}(Q) \subseteq Q''$ by functoriality. $D_n \subseteq M_n$ because $\mathbb{Z}[Q] \to \mathbb{F}_p[Q]$ maps Δ^n into $\Delta^n_{\mathbb{F}_p}$. If $d(Q) \leq 2$ then γ_2/γ_3 is cyclic and $Q'' \subseteq \gamma_5 = 1$ (see Lemma 6.3). □

Lemma 6.2 (minimal counterexample, $p = 3$)**.** *Let Q be a 3-group of least order with $D_5(Q) \neq \gamma_5(Q)$. Then Q has class 4, $Q'' \neq 1$ and $d(Q) \geq 3$; $Z(Q)$ is cyclic and $D_5(Q) = \Omega_1(Z(Q)) = \langle z \rangle$ has order 3, with $z \in Q'' \cap M_5(Q)$; the last nontrivial term $P_c(Q)$ of the lower exponent-3 central series equals $\langle z \rangle$, so Q is an immediate descendant of step size 1 of $Q/\langle z \rangle$.*

Proof. (The lemma is also formally verified, with [GT85, Thm. 3.2(c)] as a hypothesis, §4.)

If Q had class ≥ 5, then $Q/\gamma_5(Q)$ would be a smaller counterexample; for class ≤ 3, $D_5 \subseteq D_4 = \gamma_4 = 1$ (Passi: $D_4 = \gamma_4$ without 2-torsion [Pas68]). So Q has class 4, and Lemma 6.1 gives $1 \neq D_5(Q) \subseteq Q'' \cap M_5(Q)$. The rest is the argument of Lemma 5.2: $D_5(Q)$ lies in every normal N with $D_5(Q/N) = 1$, in particular in every central subgroup of order 3. □

A Lie-ring lemma.

Lemma 6.3. *Let p be odd and G a group with $|\gamma_3(G)/\gamma_4(G)| = p$ and $\gamma_2(G)/\gamma_3(G)$ generated by at most two elements. Then $[\gamma_2(G), \gamma_2(G)] \subseteq \gamma_5(G)$. In particular, G is metabelian if it has class ≤ 4.*

Proof. (The lemma is also formally verified; there the same computation is done in the group $G/\gamma_5(G)$, §4.)

The image of $[\gamma_2,\gamma_2]$ in γ_4/γ_5 is spanned by the brackets $[L_2,L_2]\subseteq L_4$ of $L=L(G)$, so it suffices to show $[L_2,L_2]=0$. If L_2 is cyclic, the alternating bracket $L_2\times L_2\to L_4$ vanishes. Otherwise $pL_3=0$, hence $pL_4=0$ ($L_4=[L_3,L_1]$), and dividing L by the ideal $pL_1\oplus pL_2\oplus 0\oplus H\oplus L_{\geq 5}$, where $H\subset L_4$ is a hyperplane missing a given nonzero bracket $[u,v]$, we get a graded $\mathbb{F}_p$-Lie algebra $M=M_1\oplus\cdots\oplus M_4$ generated by M_1 with $\dim M_2=2$, $\dim M_3=\dim M_4=1$ and $[M_2,M_2]\neq 0$. We show that this is impossible.

Write $[z,z']=\omega(z,z')w$ for $z,z'\in M_2$, $M_4=\langle w\rangle$, $M_3=\langle t\rangle$, $[x,t]=\varphi(x)w$, $[x,z]=\beta(x,z)t$ and $[x,y]=\alpha(x,y)$ for $x,y\in M_1$, $z\in M_2$. A nonzero alternating form on a plane is nondegenerate, so ω is; $\varphi\neq 0$ since $M_4=[M_1,M_3]$. Pick e with $\varphi(e)=1$ and let $K=\ker\varphi$. The Jacobi identity for $x,y\in M_1$, $z\in M_2$ reads

$$\omega(\alpha(x,y),z)=\varphi(x)\beta(y,z)-\varphi(y)\beta(x,z).$$

For $x,y\in K$ the right side vanishes, so $\alpha(K,K)=0$ and $M_2=\alpha(e,K)$; for $x=e$, $y=k\in K$ it gives $\beta(k,z)=\omega(\alpha(e,k),z)$. The Jacobi identity for $e,k_1,k_2\in K$, with $u_i=\alpha(e,k_i)$ and $\alpha(k_1,k_2)=0$, gives $\beta(k_1,u_2)-\beta(k_2,u_1)=0$, i.e. $2\,\omega(u_1,u_2)=0$. As $p\neq 2$ and the u_i run over M_2, $\omega=0$, a contradiction. □

Remark 6.4. (1) The hypothesis $p\neq 2$ is needed: by computation, SmallGroup$(128,i)$, $i=928,\dots,937$, have type $(3,2,1,1)$ and $|G''|=2$, and there are 108 such groups of order 2^8.

(2) The other two hypotheses are sharp at order 3^8: of the 904 non-metabelian 3-groups of class 4 and order 3^8, 15 have type $(3,3,1,1)$ and 889 type $(3,2,2,1)$, by computation; an explicit Lie algebra of type $(3,3,1,1)$ over $\mathbb{F}_3$ with $[M_2,M_2]\neq 0$, which after changing one constant works over every $\mathbb{F}_p$ with p odd, is given in Appendix A, where it is verified by hand; the verification is also formalized.

(3) A computational check: for all groups of orders 3^5–3^8 of class ≥ 4 the hypothesis holds in 13 077 groups and the conclusion in all of them; an independent scan of all 1 396 077 groups of order 3^8 and all groups of order 7^7 found 24 215 and 4 408 groups with $|\gamma_2/\gamma_3|=p^2$, $|\gamma_3/\gamma_4|=p$, and no violation.

(4) Compare Schneider's structure theorems for finite p-groups, p odd, with $G''\neq 1$ and $|G'/G''|=p^3$ (in particular γ_3 is then maximal in G') [Sch03], building on Blackburn and Hall, and their Lie algebra analogue [Zac08]. The algebra of Appendix A has $|M'/M''|=p^4$, just outside that range.

Proposition 6.5. *Let p be odd and Q a p-group of class 4.*

(1) *If $|Q|\leq p^7$, or $|Q|=p^8$ and $d(Q)\geq 4$, then Q is metabelian.*

(2) *If* $|Q| = p^9$ *and* $Q'' \neq 1$*, then* $d(Q) \in \{3, 4\}$*; for* $d(Q) = 4$ *the type is* $(4, 3, 1, 1)$ *or* $(4, 2, 2, 1)$*, and for* $p = 3$ *both occur (*$Q_0 \times C_3$ *with* Q_0 *of order* 3^8 *as in Remark 6.4(2)).*

Proof. (The proposition is also formally verified, except that both types occur for $p = 3$, which is a computation, see §4.)

Let (a_1, a_2, a_3, a_4) be the type. By Lemma 6.3, $Q'' \neq 1$ needs $a_3 \geq 2$ or γ_2/γ_3 with at least three generators; as γ_2/γ_3 is not cyclic, $a_2 \geq 2$, so $a_2 \geq 3$ or $a_3 \geq 2$. Moreover $a_1 \geq d(Q)$, $a_4 \geq 1$, and for $d(Q) = 2$ the group γ_2/γ_3 is cyclic. So $d(Q) \geq 3$ and $|Q| \geq p^{d(Q)+5}$, which gives (1) and the first part of (2); at p^9 and $d = 4$ the bound is attained exactly by the two types listed. At p^9 and $d = 5$ the only type with $a_2 \geq 2$ is $(5, 2, 1, 1)$, and such groups are metabelian. □

In particular the statement "every 3-group of class 4 and order 3^5, 3^6, 3^7 is metabelian" of Proposition 5.1, obtained there by computation, follows from Proposition 6.5, and so does the observation that all non-metabelian groups of class 4 and order 3^8 have rank 3.

Groups with G/γ_2 of exponent 3.

Theorem 6.6. *Let* Q *be a finite* 3*-group with* $Q/\gamma_2(Q)$ *of exponent* 3*. Then* $D_5(Q) = \gamma_5(Q)$*. More generally,* $D_5(G) = \gamma_5(G)$ *for every group* G *with* $G/\gamma_2(G)$ *finite of exponent* 3*.*

This covers non-metabelian groups of class 4 of every rank and order; for instance all 904 non-metabelian groups of class 4 and order 3^8 have $Q/\gamma_2 \cong C_3^3$, by computation.

Proof. (The proof relies on Tahara's description of D_5 [Tah81, Thm. 6.1]. The theorem is also formally verified, together with that description, §4.)

Reductions. For G as in the second statement, $\bar{G} = G/\gamma_5(G)$ is nilpotent with finite abelianization, hence finite; its lower central factors are images of tensor powers of $\bar{G}_{ab}$, so $\bar{G}$ is a 3-group, and $D_5(G) \subseteq \gamma_5(G)$ follows from $D_5(\bar{G}) = 1$ by functoriality. So let Q be a finite 3-group with $\gamma_5(Q) = 1$ and Q/γ_2 of exponent 3; we show $D_5(Q) = 1$. Tahara's Theorem 6.1 is stated for any finite group with an N-series ending at $H_5 = 1$ [Tah81, §6, p. 159], so class ≤ 3 needs no separate treatment. We write γ_3 and γ_4 additively where convenient.

Identities. (i) γ_2/γ_3, γ_3/γ_4, γ_4 are elementary abelian: they are spanned by the values of bilinear maps with one argument in Q/γ_2. (ii) $P\colon Q/\gamma_2 \to \gamma_2/\gamma_3$, $x \mapsto x^3\gamma_3$, and $\pi\colon \gamma_2/\gamma_3 \to \gamma_3/\gamma_4$, $a \mapsto a^3\gamma_4$, are homomorphisms. (iii) For $x \in Q$, $a \in \gamma_2$:

$$[x^3, a] = [x, a]^3 = [x, a^3], \qquad [a, x^3] = [a, x]^3.$$

Indeed, for $c = [x, a] \in \gamma_3$, $c^x = c[c, x]$ with $[c, x]$ central, so $[x^3, a] = c^3[c, x]^3 = c^3$ as γ_4 has exponent 3; and $[x, a^3] = c\, c^a c^{a^2} = c^3$ as $[\gamma_3, \gamma_2] = 1$. Consequently, if $a \in \gamma_2$ and $a^3 \in \gamma_4$, then $[x^3, a] = [x, a^3] = 1$.

Tahara's description. Choose bases of uniqueness: $x_1,\dots,x_s$ of Q/γ_2 such that $x_1,\dots,x_m$ span a complement of $\ker P$ and $x_{m+1},\dots,x_s$ span $\ker P$ (so $x_i^3\in\gamma_3$ for $i>m$); $x_{2k}=x_k^3$ for $k\le m$, completed to a basis $x_{21},\dots,x_{2t}$ of γ_2/γ_3. All orders $d(i),e(k),f(l)$ equal 3, and the constants b_{ik} of $x_i^3\equiv\prod_k x_{2k}^{b_{ik}}\pmod{\gamma_3}$ are $b_{ik}=\delta_{ik}$ for $i\le m$, $b_{ik}=0$ for $i>m$. By [Tah81, Thm. 6.1] (pp. 159–161), $D_5(Q)$ is generated by the elements

$$g=\sum_{i<j}u_{ij}[x_i^3,x_j]+\sum_{i,k}v_{ik}\sum_{l>k}b_{il}[x_{2l},x_{2k}]+\sum_{i\le j\le k}w_{ijk}[x_i^3,x_j,x_k]$$

for integer families (u,v,v',w,w',w'') satisfying Tahara's conditions (A)–(F) (Tahara writes g with minus signs; the conditions are homogeneous, so the subgroup is the same). With all orders 3 we use: (B) $\sum_{h>i}u_{ih}b_{hk}-\sum_{h<i}u_{hi}b_{hk}+3v_{ik}+3v'_{ik}=0$; (C) $3\mid u_{ij}$; (D) $3v_{ik}-\sum_{h\le i}w_{hii}b_{hk}-\sum_{h>i}w''_{iih}b_{hk}\equiv 0$ and, for $i<j$, $\sum_{h\le i}w_{hij}b_{hk}+\sum_{i<h\le j}w'_{ihj}b_{hk}+\sum_{h>j}w''_{ijh}b_{hk}\equiv 0\pmod 3$; (E) $\sum_k v'_{ik}d_{kl}\equiv$ a combination of u-terms and of terms $w_{gih}\alpha_l^{(gh)}$, $w_{ghi}\alpha_l^{(gh)}$, $w'_{igh}\alpha_l^{(gh)}\pmod 3$, where $x_{2k}^3\equiv\prod_l x_{3l}^{d_{kl}}$ and $[x_g^3,x_h]\equiv\prod_l x_{3l}^{\alpha_l^{(gh)}}\pmod{\gamma_4}$; (F) $\sum_i v_{ik}b_{ik}\equiv 0$ and $\sum_i v_{ik}b_{il}+\sum_i v_{il}b_{ik}\equiv 0\pmod 3$ for $k<l$. Condition (C) is Tahara's (C) for $p=3$ and $d(i)=d(j)$, where $\binom{3}{3}=1$ is not divisible by 3; for $p\ge 5$ it is empty.

$g=0$ for every admissible family. (1) Putting $b_{hk}=\delta_{hk}$ ($k\le m$) into (D) shows that every coefficient w,w',w'' whose distinguished index (the first, middle, last one respectively) is $\le m$ is divisible by 3. (2) Hence the w-part vanishes: for $i\le m$ by (1) and since γ_4 has exponent 3; for $i>m$ because $x_i^3\in\gamma_3$ and $[x_i^3,x_j,x_k]\in\gamma_5=1$. (3) Writing $u_{ij}=3u'_{ij}$ by (C), identity (iii) gives $u_{ij}[x_i^3,x_j]=u'_{ij}[x_i^3,x_j^3]$, which vanishes unless $i,j\le m$. Put $Y_k=x_k^3$ ($k\le m$). (4) The v-part equals $\sum_{k<i\le m}v_{ik}[Y_i,Y_k]$. (5) With U the skew matrix $U_{ik}=u'_{ik}$ ($i<k$), (B) gives $v'_{ik}=-v_{ik}-U_{ik}$ for $k\le m$ and $v'_{ik}=-v_{ik}$ for $k>m$; (F) gives $v_{kk}\equiv 0$, $v_{lk}\equiv -v_{kl}$ for $k,l\le m$ and $v_{kl}\equiv 0$ for $k\le m<l$; in (E) all w-terms vanish modulo 3 by (1), since for a distinguished index $g>m$ we have $\alpha^{(gh)}\equiv 0$. So (E) says that $a_i=\prod_k x_{2k}^{v'_{ik}}$ satisfies $a_i^3\in\gamma_4$. With $M=v+U$ (skew modulo 3 on indices $\le m$) we get $a_i\gamma_3=-\rho_i$, $\rho_i=\sum_{k\le m}M_{ik}Y_k\in\ker\pi$. (6) Altogether

$$g=\sum_{k<i\le m}M_{ik}[Y_i,Y_k]=\tfrac12\sum_{i\le m}[x_i^3,r_i]$$

with $r_i\in\gamma_2$ representing ρ_i (dividing by 2 is legitimate in γ_4 of exponent 3), and $[x_i^3,r_i]=[x_i,r_i^3]=1$ by (iii), since $r_i^3\in\gamma_4$ is central. So $g=1$. □

Remark 6.7. The exponent hypothesis is used in three places: in (C) (for $d(j)\ge 3d(i)$ Tahara's condition is empty and terms $u_{ij}[x_i^{d(i)},x_j^3]$ with $x_j^3\notin\gamma_2$ appear), in the choice of basis (Tahara's (D), (E) hold only modulo $(d(i),e(k))$, $(d(i),f(l))$), and in (iii). The next case, $Q/\gamma_2\cong C_{3^a}\times C_3^{s-1}$,

is settled by Theorem 6.8 below. A computational check of the theorem: 169 random non-metabelian groups of class 4 with $Q/\gamma_2 \cong C_3^3$ or C_3^4, of orders 3^{10}–3^{79}, 29 of them with $\Omega_1(Z) \cap Q'' \cap M_5 \neq 1$: identity (iii) holds and $D_5 = 1$ in all of them (truncated Magnus algebra modulo 3^{19} with the exactness certificate).

Theorem 6.8. *Let Q be a finite 3-group with $Q/\gamma_2(Q) \cong C_{3^a} \times C_3^{s-1}$ for some $a \geq 1$, i.e. with at most one invariant of Q/γ_2 larger than 3. Then $D_5(Q) = \gamma_5(Q)$; the same holds for every group G with $G/\gamma_2(G)$ of this form.*

Sketch; the case $a = 2$ is written out in Appendix B. (As for Theorem 6.6, the proof relies on [Tah81, Thm. 6.1], and the theorem is also formally verified, §4.)

For $a = 1$ this is Theorem 6.6; let $a \geq 2$, $\gamma_5(Q) = 1$, and let x_s have order 3^a modulo γ_2. Since $\Lambda^2(C_{3^a} \times C_3^{s-1})$ has exponent 3, the factors γ_2/γ_3, γ_3/γ_4, γ_4 are still elementary abelian, so identities (i)–(iii) of the proof of Theorem 6.6 hold for all $x \in Q$, and moreover $[x^{3^n}, y] = [x, y]^{3^n}$ for $n \geq 2$. The class $x_s^{3^a}\gamma_3$ does not depend on the choice of x_s, so there are two cases. If $x_s^{3^a} \notin \gamma_3$, Tahara's (B) forces $3 \mid u_{is}$, every term involving $x_s^{3^a}$ vanishes by the identity above, and the proof of Theorem 6.6 goes through. If $x_s^{3^a} \in \gamma_3$, the coefficients u_{is} are unrestricted. For $a \geq 3$ all new terms are 3^a-th powers of elements of γ_3 and vanish. For $a = 2$, (E) now gives $\sum_i [x_i^3, r_i] = -\sum_i u_{is}[x_i, x_s]^9$, so the pairing $\frac{1}{2}\sum_i [x_i^3, r_i]$ cancels, formally, only half of the new term $\sum_i u_{is}[x_i^9, x_s]$ of g; the remainder is $\frac{1}{2}[r_s^3, x_s]$ with $r_s = \prod_i x_i^{3u_{is}}$, and condition (E) for the row s, which sees u_{is} through (B) with $d(s)/d(i) = 3$, gives $r_s^3 \in \gamma_4$. So $g = 1$ for every admissible family. (This step is not void: a computation finds groups of order 3^{16} with $x_s^9 \in \gamma_3$ and $[x_i^9, x_s] \neq 1$.) Here $\alpha^{(ss)} = 0$ is read off Tahara's generators of types (V), (VI) [Tah81, pp. 150–151]. □

Consequently a minimal 3-group with $D_5 \neq \gamma_5$ has at least two invariants ≥ 9 in Q/γ_2, so $|Q/\gamma_2| \geq 3^5$. Computational checks: 3 283 random groups of class 4 with $Q/\gamma_2 \cong C_{3^a} \times C_3^{s-1}$, $3^a \in \{9, 27, 81\}$, $s = 2, 3, 4$ (for $s = 2$ all are metabelian, as they must be), 2 731 of them non-metabelian, 2 439 of these with $x_s^{3^a} \in \gamma_3$ (including, in the runs that recorded it, 296 with $[x_i^9, x_s] \neq 1$); of the 8 629 candidate central elements z none lies in D_5. The 20 groups in which the subgroup of candidates has order $> 3^5$ were skipped.

The p-covering group. For a p-group P of p-class c let P^* be its p-covering group, $M \leq Z(P^*)$ the multiplier (elementary abelian, $P^*/M \cong P$) and $N = P_{c+1}(P^*) \leq M$ the nucleus [OBr90]. The immediate descendants of step size 1 are the $Q = P^*/U$, $U < M$ of index p with $N \not\leq U$, and then $\langle z \rangle = M/U$ is central of order p with $Q/\langle z \rangle \cong P$.

Lemma 6.9 (a filter on parents)**.** *If one of $M \cap P^{*\prime\prime}$, $M \cap M_5(P^*)$, N and, when P has class 3, $\gamma_4(P^*)$, lies in $\gamma_5(P^*)$, then no immediate descendant Q of step size 1 of P is of class 4 with $z \in Q'' \cap M_5(Q)$.*

Proof. (The lemma is also formally verified, in a more general form, §4.)

γ_n, derived subgroups and the Jennings subgroups $M_n = \prod_{ip^j \geq n} \gamma_i^{p^j}$ pass to quotients: $X(P^*/U) = X(P^*)U/U$. Class 4 needs $\gamma_5(P^*) \leq U$ (and $\gamma_4(P^*) \not\leq U$ if P has class 3); $z \in Q''$ needs $M \cap P^{*\prime\prime} \not\leq U$; $z \in M_5(Q)$ needs $M \cap M_5(P^*) \not\leq U$; and $N \not\leq U$. □

Theorem 6.10. *Let p be odd and P a finite p-group of class 3 with $|\gamma_2(P)/\gamma_3(P)| = p^2$ and $|\gamma_3(P)| = p$. Then $P^{*\prime\prime} = 1$. In particular every immediate descendant of step size 1 of P is metabelian, and P is rejected by Lemma 6.9.*

Proof. (The theorem is also formally verified, in a more general form, §4.)

$P'' \subseteq \gamma_4(P) = 1$, so $P^{*\prime\prime} \leq M$, and as the subgroups U of index p in M intersect trivially it suffices to show $Q'' = 1$ for $Q = P^*/U$. Here $\gamma_4(Q) \leq \langle z \rangle$. If $z \notin \gamma_2(Q)$, then $\gamma_2(Q) \cong \gamma_2(P)$ is abelian. If $z \in \gamma_2(Q) \setminus \gamma_4(Q)$, then $\gamma_4(Q) = 1$ and $Q'' \subseteq \gamma_4(Q) = 1$. If $z \in \gamma_4(Q)$, then Q has the same factors γ_2/γ_3, γ_3/γ_4 as P, and Lemma 6.3 gives $Q'' \subseteq \gamma_5(Q) = 1$. (Lemma 6.3 does not apply to P^* itself, whose lower central factors are larger.) □

Remark 6.11.

(1) For $p = 2$ the theorem fails, as a computation shows: SmallGroup$(64, 138)$ has type $(3, 2, 1)$, while its 2-covering group has order 2^{13}, class 4 and $|P^{*\prime\prime}| = 2$; likewise SmallGroup$(128, i)$, $i = 522, 599, 621, 1755$, and 12 groups of order 2^8.

(2) A computational check: the hypothesis holds for 13 315 groups of order 3^8 and 2 754 of order 7^7, and $P^{*\prime\prime} = 1$ in all of them.

(3) The theorem does not say that such P are terminal. Of the 7 824 groups of order 3^8, type $(5, 2, 1)$ and rank 5 (SmallGroup$(3^8, i)$, $1\,387\,558 \leq i \leq 1\,395\,381$), a computation finds that 190 have nucleus of order 3, e.g. SmallGroup$(81, 9) \times 3^{1+2}_+ \times C_3$, and that all of their descendants are metabelian of type $(5, 2, 1, 1)$. Theorem 6.12 below says exactly which groups of type $(d, 2, 1)$ with P/γ_2 elementary abelian are nonterminal, and on this block it selects exactly the 190 (by computation). Its conditions (N1)–(N3) and (E) alone hold for the 190 and also for 573 terminal groups; the Jacobi condition (J) separates the two.

Theorem 6.12. *Let p be odd and P a finite p-group of type $(d, 2, 1)$ with $P/\gamma_2(P)$ elementary abelian, and write $[a, b] = a^{-1}b^{-1}ab$. Then P is nonterminal (it has an immediate descendant, equivalently its nucleus is nontrivial) if and only if* (N1) *$|Z(P) \cap \gamma_2(P)| = p^2$,* (N2) *$\gamma_2(P)$ is elementary abelian,* (N3) *$P^p \leq Z(P)$, and* (N4) *the linear system* (E)+(J) *below over $\mathbb{F}_p$ has a solution with $c = 1$.*

To state (N4), assume (N1)–(N3) and choose a generator w of $\gamma_3(P)$, elements $v \in Z(P) \cap \gamma_2(P) \setminus \gamma_3(P)$ and $u \in \gamma_2(P) \setminus Z(P)$, and generators

$x_1, \dots, x_d$ of P. Then $\gamma_2(P) = \langle u\rangle \times \langle v\rangle \times \langle w\rangle$, $x_i^p \in \langle v, w\rangle$, and

$$[x_i, x_j] = u^{\omega_1(i,j)} v^{\omega_2(i,j)} w^{\omega_3(i,j)}, \qquad x_i^p = v^{\alpha_i} w^{\beta_i}, \qquad [u, x_i] = w^{f_i},$$

with alternating ω_r and $f \neq 0$. The unknowns are $c, \mu_u, \mu_v \in \mathbb{F}_p$ and $\eta, \sigma \in \mathbb{F}_p^d$; let $\varepsilon_3 = 1$ and $\varepsilon_p = 0$ for $p \geq 5$: for all $1 \leq i, j \leq d$, including $i = j$,

$$\text{(E)} \qquad \alpha_j \eta_i + c\,\beta_j f_i + \mu_u\,\omega_1(i,j) + \mu_v\,\omega_2(i,j) + \varepsilon_p\, c\, f_j^2\,\omega_1(i,j) = 0,$$

and for all $1 \leq i < j < k \leq d$, with (a, b, e) running over the cyclic shifts of (i, j, k),

$$\text{(J)} \qquad \textstyle\sum_{(a,b,e)} \big(\omega_1(a,b)\,\sigma_e + \omega_2(a,b)\,\eta_e + c\,\omega_3(a,b)\,f_e\big) = 0.$$

Sketch; the full proof is in Appendix C. (The theorem is also formally verified, with "nonterminal" in the form of Lemma C.1, see §4; Remark 6.13(1) reports a computational check.)

P has p-class 3, so it is nonterminal if and only if there is a group Q with a central subgroup $\langle z\rangle$ of order p, $Q/\langle z\rangle \cong P$ and $\gamma_4(Q) \neq 1$. In such Q we have $\gamma_4(Q) = \langle z\rangle$, $\gamma_5(Q) = 1$, $Q'' = 1$ by Lemma 6.3, and $\gamma_3(Q)$ of exponent p; (N1)–(N3) follow. Write $A = \gamma_2(Q)$ additively and $D_x a = [a, x]$: the D_x commute on A and products of three of them vanish. Then the Hall–Witt identity becomes $[x, y, t] + [y, t, x] + [t, x, y] = 0$ exactly, and its z-component on generators is (J). Comparing $(x_j^p)^{x_i}$ with $(x_j[x_j, x_i])^p$ gives (E). The unknowns are the exponents of z in $[v, x_i] = z^{\eta_i}$, $[u, x_i] = w^{f_i} z^{\sigma_i}$, $[w, x_i] = z^{c f_i}$, $u^p = z^{\mu_u}$, $v^p = z^{\mu_v}$. Conversely, a solution defines a pc-presentation on $x_1, \dots, x_d, u, v, w, z$. Its consistency test words differ in the exponent of z by $-J$, $\pm E$ or 0, so it is consistent, and the group it defines has $\gamma_4 = \langle z\rangle \neq 1$. □

Remark 6.13. (1) A computational check. For the 7 824 groups of Remark 6.11(3) and for all groups of order p^7, type $(4, 2, 1)$ and rank 4 (i.e. with P/γ_2 elementary) with $p = 3, 5, 7$ (932, 1 460, 2 232 groups), the nucleus was computed in two ways: with ANUPQ, and with GAP's own p-quotient of $F/[R, F]R^p$. The two agree, and $N \neq 1$ exactly for the groups satisfying (N1)–(N4): 190, 102, 110, 118 groups. For each of these, the pc-presentation of the proof built from the solution was checked to be consistent, with $\gamma_4 \neq 1$, and to map onto P; so for these groups sufficiency does not rest on the consistency test. Without the term $c\,\omega_3 f$ in (J) the criterion gives wrong answers both ways: on the block it accepts 295 terminal groups and rejects 13 nonterminal ones. Code: `compute/magnus`.

(2) The system is linear and homogeneous in $(c, \eta, \sigma, \mu_u, \mu_v)$, so (N4) is the solvability of an inhomogeneous linear system. The term with ε_p is $\binom{p}{3}[x_j, x_i, x_j, x_j]$, the only place where $p = 3$ differs from $p \geq 5$. Whether (N4) holds does not depend on the choices made, since it is equivalent to nonterminality.

(3) The criterion assumes P/γ_2 elementary abelian, i.e. $d(P) = d$; this implies that P has p-class 3, but not conversely. It covers all groups of order 3^8, type $(5, 2, 1)$ and rank 5, i.e. all possible parents in Theorem 6.14, but not all groups of type $(5, 2, 1)$: for instance $M \times (C_9 \rtimes C_9)$, with M of maximal class and order 81 and $C_9 \rtimes C_9$ non-abelian, has order 3^8, type $(5, 2, 1)$ and rank 4. Groups of type $(d, 2, 1)$ with P/γ_2 not elementary are not treated here.

Orders 3^8 and 3^9.

Theorem 6.14 (Computer-assisted)**.** *$D_5(Q) = \gamma_5(Q)$ for every 3-group Q of order at most 3^9. Hence $D_5(G)/\gamma_5(G)$ has no 3-torsion for every finite group G whose Sylow 3-subgroup has order at most 3^9.*

Proof. By Theorem 5.4(3) a counterexample Q of least order has order 3^8 or 3^9, and Lemma 6.2 applies to it; for order 3^9 we may use the result for order 3^8. The deduction for G is as in Theorem 5.4.

Order 3^8. Every group is in the SmallGroups library (package `sglppow` [VLE]). Of the 1 396 077 groups, 904 are non-metabelian of class 4; all have rank 3 (as they must, Proposition 6.5), $|Z| = |Q''| = |\gamma_4| = 3$ and $Q/\gamma_2 \cong C_3^3$, types $(3, 2, 2, 1)$ (889) and $(3, 3, 1, 1)$ (15), by a computation over the library. So Theorem 6.6 gives $D_5 = 1$ for all of them. Independently, 699 of them have $z \in Q'' \cap M_5$, and for each the truncated Magnus algebra (121 monomials, modulo 3^{19}, with the exactness certificate $3^{18}\Delta \subseteq \Delta^5 + 3^{19}$) gives $z \notin D_5$.

Order 3^9. Let Q be a counterexample and $P = Q/\langle z\rangle$, of order 3^8 and class 3 or 4. By Proposition 6.5(2), $d(Q) \in \{3, 4\}$; the case $d(Q) = 5$, type $(5, 2, 1, 1)$, is excluded by hand (alternatively, by Theorem 6.10: all 7 824 possible parents are rejected by Lemma 6.9, as a full run confirms). For $d \in \{3, 4\}$ we use computation: of the 771 597 groups P of order 3^8, rank 3 or 4 and class 3 or 4, 3 986 pass the filter of Lemma 6.9; ANUPQ lists their 305 583 immediate descendants of step size 1; 64 362 have class 4, $Q'' \neq 1$ and cyclic centre, and 62 245 of these have $z \in Q'' \cap M_5$. For each of the 62 245 the truncated Magnus algebra gives $z \notin D_5$, exactly. The presentations given to it were checked independently with ANUPQ (order 3^9, class 4, z central of order 3 in $Q'' \cap M_5$). The filter was validated against full expansion on a range and a random sample of 1500 parents, without loss of a candidate. The descendant step was repeated with an ANUPQ in which the defects found in an audit of the package are fixed (`https://ispacex.github.io/gap/`; two of them give wrong descendants for inputs not met here): for each of the 3 986 parents the numbers of descendants, of groups of class 4 with $Q'' \neq 1$ and cyclic centre, and of candidates are the same, and the 62 245 presentations are identical. □

Cost and data. The order-3^9 run took 3 h 21 min on 4 cores; the filter cut the time per parent from about 350 to 27 ms. Code: `compute/magnus`.

7. Open questions

(1) Question 1.1; equivalently, is $D_5(Q) = 1$ for all finite 2-groups of class 3? A counterexample must have $D_4(Q) \neq 1$ and order $\geq 2^{12}$ (Proposition 5.10); at order 2^{12} it is a candidate of rank ≤ 8 (after Proposition 5.10). Already $D_4 \neq \gamma_4$ needs order $\geq 2^{11}$ (§5, after the sections of Rips's group), while the smallest known example, Rips's group, has order 2^{38}. Using the truncated Magnus algebra $(\mathbb{Z}[G]/\Delta^n \cong T_{<n}(\mathbb{Z}^k)/I_n$ for a k-generator presentation, of rank $(k^n-1)/(k-1)$ independent of $|G|$) we checked the class-3 quotients of the three-relator Rips-type group of [BM23, §1.2] and of 13 variants (orders 2^{40}–2^{107}): $D_5 = 1$ in all of them, including three, of orders 2^{84}, 2^{90} and 2^{107}, with $D_4 \neq 1$. Two independent implementations agree on these results, and the exactness certificate holds in all cases, so $D_4 \neq 1$ and $D_5 = 1$ are proved, not only modulo 2^{30}. The method (code: `https://gitlab.com/ispacex/cflint-core`) was also checked against the lattice computation on 30 435 groups of order $\leq 3^7$ without discrepancies, and is 200–400 times faster at order 3^7.

(2) Is there 3-torsion in $D_5(G)/\gamma_5(G)$ for some G, i.e. is Tahara's bound 6 attained? ("Rips's abyss" in Mikhailov's talk [Mik13].) Tahara expected the bound 6 to be the best one [Tah81, p. 142]. At the end of the paper he announced a separate paper [Tah81, p. 173], which would show $D_5(G) = \gamma_5(G) = 1$ for his 2-groups with $D_4 \neq \gamma_4$ [Tah77b] and construct counterexamples to $D_5(G) = \gamma_5(G)$. The announcement does not say that their quotient D_5/γ_5 would have 3-torsion, and $D_5 \neq \gamma_5$ with 2-torsion is known anyway: $2 \mid s(5)$ in the notation of [BM23, §1]. We have not been able to locate this paper (the conference report [THSH81] of 1981 is on other topics), and we could not obtain his surveys in Bull. Aichi Univ. Educ. (1981, 1982). According to its zbMATH review, Tahara's survey of 2003 [Tah03] reports Gupta's claim that the exponent of D_n/γ_n is at most 2. Bartholdi and Mikhailov [BM23, §1] state the bound as $s(5) \in \{2, 6\}$, i.e. open. By the results of this paper, there is no 3-torsion when the Sylow 3-subgroup has order $\leq 3^9$ (in D_n/γ_n for every n, see below), and $D_5 = \gamma_5$ when G/γ_2 is finite of exponent 3 (§6); by [Gup91] a finite metabelian group has no 3-torsion in D_n/γ_n, so an example must be non-metabelian. A minimal 3-group with $D_5 \neq \gamma_5$ has order $\geq 3^{10}$, class 4, cyclic centre, and at least two invariants ≥ 9 in Q/γ_2 (Theorem 6.8). Not covered by Theorems 6.6 and 6.8: Q/γ_2 with two invariants ≥ 9, e.g. $C_9 \times C_9 \times C_3$, where γ_2/γ_3 may have exponent 9. For every n, a 3-group with $D_n \neq \gamma_n$ has order $\geq 3^{10}$; the smallest known example, a quotient of the Bartholdi–Mikhailov group, has order 3^{493} (§5). Let Q be a 3-group of least order with $D_n \neq \gamma_n$. Then $\gamma_n(Q) = 1$ (pass to Q/γ_n),

so Q has class $c < n$. Every proper quotient has $D_n = \gamma_n = 1$, so $D_n(Q) \neq 1$ lies in every nontrivial normal subgroup; hence the centre is cyclic and its socle $\langle z \rangle = D_n(Q)$ lies in $D_{c+1}(Q)$, in the Jennings subgroup $M_{c+1}(Q)$ and, by [Gup91] applied to Q/Q'', in Q''. Class ≤ 3 is impossible ($D_4 = \gamma_4$ in a 3-group [Pas68]), and class 4 gives $D_n \subseteq D_5 = \gamma_5 = 1$ by Theorem 6.14. Of the groups of order 3^6, 3^7, 3^8 of class ≥ 5, 1035 satisfy these conditions, and all have $z \notin D_{c+1}$, by an exact computation in the truncated Magnus algebra (code: `compute/magnus`). At order 3^9 a least-order Q of class ≥ 5 is an immediate descendant of step size 1 of $P = Q/\langle z \rangle$, since the last term of its lower exponent-3 central series is nontrivial, central and of exponent 3, hence lies in $\Omega_1(Z(Q)) = \langle z \rangle$ and equals it; P has class $\geq c - 1 \geq 4$ and rank $d(Q) \leq 5$, as $d(Q) + c - 1 \leq 9$. Of the 66 569 such P, 4 627 pass a necessary test on the p-cover (with $Q = P^*/U$, each of $z \in Q''$, $z \in M_6(Q)$, $\gamma_5(Q) \neq 1$ requires a subgroup of the multiplier not contained in U, and so does Q being a descendant: the nucleus is not in U); ANUPQ lists their 65 398 immediate descendants, 27 547 of them satisfy the conditions above, and all have $z \notin D_{c+1}$, by the same exact computation; the descendant step repeated with the corrected ANUPQ gives the same 27 547 presentations. (The nonterminal parents of type $(5, 2, 1)$ are characterised exactly by Theorem 6.12; this is not needed for Theorem 6.14.)

(3) The group analogue of Theorem C of [IMP26]: is $D_{2c-1}(G) = 1$ for every metabelian group of class $\leq c$? For $c = 3$ this is Question 1.1. The question reduces to finite metabelian p-groups (a finite nilpotent quotient is the product of its Sylow subgroups); for p odd the answer is yes for every c by [Gup91] (for $c \leq p + 1$ already by [GT85, Thm. 3.2]). So only $p = 2$, $c \geq 3$ is open. The plateau group of [IMP26, §3.3] is metabelian of class 4 with $D_6 \neq 1$ (by computation), so $2c - 2$ is attained, as for Lie rings. Its distinguished element w is not in D_7, and since $D_6 = \langle w \rangle$ [IMP26], $D_7 = 1$ (computed from the dense Howell form modulo 2^{32} of the ideal of relations in the truncated Magnus algebra of rank 55 986, with the exactness certificate): this group is not a counterexample, and the analogue, if true, is sharp at $c = 4$. A first search for a counterexample at $c = 4$ reduces to metabelian quotients of class 4 with γ_4 finite cyclic, testing its element of order 2 (this reduction uses $D_5^2 \subseteq G''\gamma_5$ [MP09, Thm. 2.28] and is proved by hand in the documentation of the code, `https://gitlab.com/ispacex/cflint-fast-core`, `docs/analysis.md`, §6.6); about 170 such groups (quotients and perturbations of the plateau group, group versions of the Lie ring L_1 of [IMP26, Ex. 3.2], random presentations on 4 generators) all have $D_7 \cap \gamma_4 = 1$ by computation. For $c = 5$, the plateau group shifted up one class (five generators) has $D_7 \neq 1$ but $D_8 \cap \gamma_5 = 1$

(by a certified computation), so for metabelian groups of class 5 at least $D_7 \neq 1$ is attained, one short of $2c-2$; an elementary pairing lemma indicates that the construction would need the exponents of the relations of [IMP26, (3.12)] quadrupled rather than doubled at class 5 (the lemma is proved by hand in the same documentation, §6.7), and in our computer experiments with relations modified in this way the distinguished element w becomes trivial. Code and data: `https://gitlab.com/ispacex/cflint-fast-core` (`docs/analysis.md`, §§6.5–6.7).

Appendix A. A Lie algebra of type $(3,3,1,1)$

This is the algebra of Remark 6.4(2). Let $M = M_1 \oplus M_2 \oplus M_3 \oplus M_4$ over $\mathbb{F}_3$ with bases $M_1 = \langle e, k_1, k_2\rangle$, $M_2 = \langle u_1, u_2, \rho\rangle$, $M_3 = \langle t\rangle$, $M_4 = \langle w\rangle$, and the alternating bracket whose nonzero values on pairs of basis elements are, up to antisymmetry,

$$= u_1, \quad [e,k_2] = u_2, \quad [k_1,k_2] = \rho,$$
$$[k_1,u_2] = t, \quad [k_2,u_1] = -t, \quad [e,\rho] = -t, \quad [e,t] = w, \quad [u_1,u_2] = w.$$

It is graded, generated by M_1, of type $(3,3,1,1)$, and $[M_2,M_2] = M_4 \neq 0$.

Jacobi identity. This is also formally verified, with $[e,\rho] = 2t$, which is the same over $\mathbb{F}_3$ (§4). The Jacobiator $J(x,y,z) = [[x,y],z]+[[y,z],x]+[[z,x],y]$ is alternating and trilinear and vanishes in degrees ≥ 5, so it suffices to check triples of distinct basis elements of total degree 3 or 4. Degree 3: $J(e,k_1,k_2) = [u_1,k_2] + [\rho,e] + [k_1,u_2] = t+t+t = 0$, since $3 = 0$ in $\mathbb{F}_3$. Degree 4, triples (x,y,z) with $x,y \in M_1$, $z \in M_2$: as $[k_i,t] = 0$ and $[\rho,M_2] = 0$, $J(k_1,k_2,z) = [\rho,z]-[k_1,[k_2,z]]+[k_2,[k_1,z]] = 0$ and $J(e,k_i,z) = [u_i,z]-[e,[k_i,z]]+[k_i,[e,z]] = [u_i,z]-[e,[k_i,z]]$; the last two terms are both w for $(i,z) = (1,u_2)$, both $-w$ for $(i,z) = (2,u_1)$, and both 0 otherwise. (A machine check of all triples agrees.)

In the notation of the proof of Lemma 6.3, $\varphi(e) = 1$, $K = \langle k_1,k_2\rangle$, $u_i = \alpha(e,k_i)$, and $\alpha(k_1,k_2) = \rho \neq 0$ spans the radical of ω: with $\dim M_2 = 3$ the form ω may be degenerate, and the step $\alpha(K,K) = 0$ of that proof fails. The Jacobi identity for (e,k_1,k_2), which there gives $2\,\omega(u_1,u_2) = 0$, here reads $2\,\omega(u_1,u_2) = \beta(e,\rho)$, both sides being -1; with these constants it holds only because $3t = 0$. Taking $[e,\rho] = 2t$ instead (the same over $\mathbb{F}_3$) gives such an algebra over $\mathbb{F}_p$ for every odd p: then $J(e,k_1,k_2) = t-2t+t = 0$ and both sides above equal 2, while the degree-4 triples are unaffected as $[k_i,t] = 0$. The Jacobi identity over any commutative ring, and the type and $[M_2,M_2] \neq 0$ over any field, are also formally verified (§4); a machine check of all triples for $p = 3,5,7,11$ agrees.

Appendix B. The case $a = 2$ of Theorem 6.8

We write out the case $a = 2$, $x_s^9 \in \gamma_3$, of the proof of Theorem 6.8, the only case in which the new terms do not vanish by themselves. The other cases are as in the sketch.

Setting. Q is a finite 3-group of class ≤ 4, $Q/\gamma_2 \cong C_9 \times C_3^{s-1}$ with $s \ge 2$, and $\gamma_i = \gamma_i(Q)$. Then γ_3 is abelian, γ_4 is central, $[\gamma_2, \gamma_3] = 1$, $[\gamma_2, \gamma_2] \subseteq \gamma_4$, and for $x \in Q$ the maps $c \mapsto [c, x]$ and $c \mapsto [x, c]$ are homomorphisms $\gamma_3 \to \gamma_4$ trivial on γ_4.

Identities. (a) γ_2/γ_3, γ_3/γ_4 and γ_4 are elementary abelian. Indeed, for a basis $\bar{x}_1, \dots, \bar{x}_s$ of Q/γ_2 with $\bar{x}_s$ of order 9, γ_2/γ_3 is spanned by the $[x_i, x_j]\gamma_3$, $i < j$, where $i < s$ and $[x_i, x_j]^3 \equiv [x_i^3, x_j] \equiv 1 \pmod{\gamma_3}$ as $x_i^3 \in \gamma_2$; the other two factors are spanned by the values of bilinear maps whose first argument lies in the previous factor. Hence the proofs of the identities of Theorem 6.6 apply: $\pi\colon \gamma_2/\gamma_3 \to \gamma_3/\gamma_4$, $a \mapsto a^3\gamma_4$, is a homomorphism and $(ab)^3 = a^3b^3$ for $a, b \in \gamma_2$; and identity (iii) holds for all $x \in Q$. Moreover, by the Hall–Petrescu formula $(xy)^3 \equiv x^3y^3 \pmod{\gamma_3}$, so $x \mapsto x^3\gamma_3$ is a homomorphism P from $\Omega = \{x \in Q : x^3 \in \gamma_2\}$ to γ_2/γ_3, trivial on γ_2. (b) $[x^9, y] = [x, y]^9$ for all $x, y \in Q$. Let $c = [x, y]$. Since $[c, x] \in \gamma_3$ commutes with c and $[c, x, x]$ is central, $[x^3, y] = c^{x^2}c^x c = c^3 f$ with $f = [c, x]^3[c, x, x] \in \gamma_4$. Put $Z = x^3$ and $d = [Z, y] \in \gamma_2$. By (iii), $[d, Z] = [d, x]^3 \in \gamma_4$ is central, so $[x^9, y] = [Z^3, y] = d^{Z^2}d^Z d = d^3[d, Z]^3 = d^3 = c^9 f^3 = c^9$.

Basis. Let $W = \Omega/\gamma_2 \cong \mathbb{F}_3^s$, $\bar{P}\colon W \to \gamma_2/\gamma_3$ the map induced by P, $m = \operatorname{rk} \bar{P}$, and $\bar{z} = x_s^3\gamma_2 \in W$; then $\bar{P}(\bar{z}) = x_s^9\gamma_3 = 0$. Choose $x_1, \dots, x_{s-1} \in \Omega$ such that $\bar{P}(\bar{x}_1), \dots, \bar{P}(\bar{x}_m)$ is a basis of $\operatorname{im} \bar{P}$ and $\bar{x}_{m+1}, \dots, \bar{x}_{s-1}, \bar{z}$ is a basis of $\ker \bar{P}$. Then $\bar{x}_1, \dots, \bar{x}_{s-1}$ are independent modulo $\langle \bar{x}_s \rangle \cap W = \langle \bar{z} \rangle$, and $3^{s-1} \cdot 9 = |Q/\gamma_2|$, so $x_1, \dots, x_s$ is a basis of uniqueness of Q/γ_2 with $d(i) = 3$ for $i < s$ and $d(s) = 9$. Put $Y_i = x_i^3$ for $i \le m$; these are independent modulo γ_3, and we take $x_{2k} = Y_k$ for $k \le m$, completed to a basis of γ_2/γ_3, and all $e(k) = f(l) = 3$ by (a). Then $b_{ik} = \delta_{ik}$ for $i \le m$, $b_{ik} = 0$ for $m < i \le s$ (as $x_i^3 \in \gamma_3$ for $m < i < s$ and $x_s^9 \in \gamma_3$), and $\zeta := x_s^9\gamma_4 = \sum_l c_{sl}\bar{x}_{3l}$.

Tahara's conditions. By [Tah81, Thm. 6.1], $D_5(Q)$ is generated by the elements

$$g = \sum_{i<j} u_{ij} \tfrac{d(j)}{d(i)} [x_i^{d(i)}, x_j] + \sum_{i,k} v_{ik} \sum_{l>k} b_{il}[x_{2l}, x_{2k}] + \sum_{i \le j \le k} w_{ijk}[x_i^{d(i)}, x_j, x_k]$$

over the integer families satisfying (A)–(F) (signs as in the proof of Theorem 6.6). With the orders above we use: (A) $w_{iii} = 0$; (B) for $i < s$, $\sum_{i<h\le s} u_{ih}b_{hk} - \sum_{h<i} u_{hi}b_{hk} + 3v_{ik} + 3v'_{ik} = 0$, and for $i = s$, $-3\sum_{h<s} u_{hs}b_{hk} + 9v_{sk} + 3v'_{sk} = 0$; (C) $3 \mid u_{ij}$ for $i < j < s$ (for $j = s$ it is empty, modulo $d(i) = 3$: in the part of Tahara's (C) with $\binom{d(i)}{3}$ and $\binom{d(i)}{2}$ the u-term carries the factor $d(s)/d(i) = 3$ and $\binom{3}{2} = 3$, and in the part with $\binom{d(s)}{3}$ and $\binom{d(s)}{2}$, 3 divides $\binom{9}{3}$ and $\binom{9}{2}$); (D) and (F) exactly as in the

proof of Theorem 6.6; and (E), modulo 3,

$$\sum_k v'_{ik}d_{kl} \equiv \sum_{h>i} u_{ih}c_{hl} - \sum_{h<i} u_{hi}\tfrac{d(i)}{d(h)}\big(\alpha_l^{(hi)} + c_{hl}\big) - E^w_{il},$$

where E^w_{il} is a sum of products of a coefficient $w_{\dots}$, $w'_{\dots}$ or $w''_{\dots}$ with distinguished index g and some $\alpha_l^{(gh)}$, $g \le h$; here $\alpha^{(gg)} = 0$, as read off Tahara's generators of types (V), (VI) [Tah81, pp. 150–151].

$g = 1$ *for every admissible family.* (1) As in steps (1), (2) of the proof of Theorem 6.6, (D) with $b_{hk} = \delta_{hk}$ $(k \le m)$ shows that 3 divides every w-coefficient whose distinguished index is $\le m$. Hence the w-part of g vanishes (for $i \le m$ by this, for $m < i < s$ because $x_i^3 \in \gamma_3$, and for $i = s$ only $w_{sss} = 0$ occurs), and $E^w_{il} \equiv 0$ (for $g \le m$ by this, for $m < g < s$ because $[x_g^3, x_h] \in \gamma_4$, and for $g = s$ because then $h = s$). (2) The u-part. For $i < j < s$, as in step (3) there, $u_{ij}[x_i^3, x_j] = u'_{ij}[Y_i, Y_j]$ with $u_{ij} = 3u'_{ij}$, and this is 1 unless $i, j \le m$. For $j = s$ the term is $[x_i^3, x_s]^{3u_{is}} = [x_i^9, x_s]^{u_{is}}$, since $[x_i^3, x_s]^3 = (c^3 f)^3 = c^9$ in the notation of (b); it is 1 for $m < i < s$, where $x_i^9 \in \gamma_4$. (3) The v-part is $\sum_{k<i\le m} v_{ik}[Y_i, Y_k]$. (4) For $i \le m$, (B) and (F) give, as in step (5) there (the term $u_{is}b_{sk}$ is 0), $v'_{ik} = -v_{ik} - U_{ik}$ for $k \le m$ and $v'_{ik} = -v_{ik}$ for $k > m$, with U the skew matrix $U_{ik} = u'_{ik}$ $(i < k)$, and $v_{kk} \equiv 0$, $v_{lk} \equiv -v_{kl}$ for $k, l \le m$, $v_{kl} \equiv 0$ for $k \le m < l$. For $i = s$, (B) gives $v'_{sk} = u_{ks} - 3v_{sk}$ for $k \le m$ and $v'_{sk} = -3v_{sk}$ for $k > m$. (5) Put $a_i = \prod_k x_{2k}^{v'_{ik}} \in \gamma_2$, so that the left side of (E) is the coordinate vector of $\pi(a_i\gamma_3)$. For $i \le m$ the terms with $h < i$ vanish modulo 3 by (C), and of those with $h > i$ only $h = s$ survives; so $\pi(a_i) = u_{is}\zeta$. For $i = s$ every u-term carries the factor $d(s)/d(h) = 3$, so $\pi(a_s) = 0$. Let $M = v + U$ on indices $\le m$ (skew with zero diagonal modulo 3), $r_i = \prod_{k\le m} Y_k^{M_{ik}}$ for $i \le m$, and $r_s = \prod_{i\le m} Y_i^{u_{is}}$. By (4), $a_i \equiv r_i^{-1}$ and $a_s \equiv r_s \pmod{\gamma_3}$; therefore

$$r_i^3 \equiv x_s^{-9u_{is}} \pmod{\gamma_4} \quad (i \le m), \qquad r_s^3 \in \gamma_4.$$

(6) Writing γ_4 additively (exponent 3, so $\frac12$ makes sense),

$$g = \sum_{k<i\le m} M_{ik}[Y_i, Y_k] + \sum_{i\le m} u_{is}[x_i^9, x_s] = \tfrac12 \sum_{i\le m} [Y_i, r_i] + \sum_{i\le m} u_{is}[x_i^9, x_s].$$

By identity (iii) and (5), $[Y_i, r_i] = [x_i^3, r_i] = [x_i, r_i^3] = [x_i, x_s^9]^{-u_{is}}$, and by (b), $[x_i, x_s^9] = [x_s, x_i]^{-9} = [x_i, x_s]^9 = [x_i^9, x_s]$. So, formally, the pairing cancels only half of the new term (in γ_4, of exponent 3, $\frac12 = -1$):

$$g = \big(1 - \tfrac12\big) \sum_{i\le m} u_{is}[x_i^9, x_s] = \tfrac12 \Big[\prod_{i\le m} x_i^{9u_{is}},\, x_s\Big] = \tfrac12\,[r_s^3, x_s],$$

since $x_i^9 = Y_i^3 \in \gamma_3$ and $\prod_i Y_i^{3u_{is}} = r_s^3$ by (a). By (5), $r_s^3 \in \gamma_4$ is central, so $g = 1$. □

The last step is where the row s of (E) is needed: it sees u_{is} through v'_{sk} in (B), where $d(s)/d(i) = 3$. The terms $[x_i^9, x_s]$ need not vanish (see

the sketch). The whole computation is also part of the Lean proof of Theorem 6.8 (`ExpThreePlus.lean`, `Setup8.gen_eq_one`).

Appendix C. Proof of Theorem 6.12

Throughout, p is odd, P is as in Theorem 6.12, $V = P/\gamma_2(P) \cong \mathbb{F}_p^d$, $[a,b] = a^{-1}b^{-1}ab$, $[a,b,c] = [[a,b],c]$, and P_i is the lower exponent-p central series. Steps 2, 4 and 5 are also formally verified (Step 5 with a different construction of Q, up to the final appeal to Lemma C.1), see §4.

Step 1: p-class. $P_2 = \gamma_2 P^p = \gamma_2$, as P/γ_2 is elementary. For $a, b \in P$, $[a,b]^p \equiv [a^p, b] \equiv 1 \pmod{\gamma_3}$ since $a^p \in \gamma_2$; so γ_2/γ_3 is elementary, $P_3 = [\gamma_2, P]\gamma_2^p = \gamma_3$ and $P_4 = \gamma_4\gamma_3^p = 1$. Hence P has p-class 3, its nucleus is $N = P_4(P^*)$, and P is nonterminal if and only if $N \neq 1$, if and only if P has an immediate descendant of step size 1 [OBr90].

Lemma C.1. *P is nonterminal if and only if there is a group Q with a central subgroup $\langle z\rangle$ of order p such that $Q/\langle z\rangle \cong P$ and $\gamma_4(Q) \neq 1$. For every immediate descendant Q of step size 1, $P_4(Q) = \gamma_4(Q) = \langle z\rangle$, $\gamma_5(Q) = 1$, $Q'' = 1$, and $\gamma_3(Q)$ is elementary abelian of order p^2.*

Proof. Let Q be an immediate descendant of step size 1: $P_4(Q) = \langle z\rangle$ has order p and $Q/\langle z\rangle \cong P$. As $P_3(Q)$ maps onto $P_3(P) = \gamma_3(P)$, $P_3(Q) = \gamma_3(Q)\langle z\rangle$ and $P_4(Q) = \gamma_4(Q)\gamma_3(Q)^p$; also $\gamma_5(Q) \le P_5(Q) = 1$. For $\ell \in \gamma_2(Q)$ and $x \in Q$, $[\ell^p, x] = [\ell, x]^p$ since $[\ell, x, \ell] \in \gamma_5(Q) = 1$, and $\ell^p \in P_3(Q)$, so $[\ell,x]^p \in \gamma_4(Q)$. Hence $\gamma_3(Q)/\gamma_4(Q)$ has exponent p and $\langle z\rangle = P_4(Q) = \gamma_4(Q)$. So Q has class 4, its factors γ_2/γ_3 and γ_3/γ_4 are those of P, and Lemma 6.3 gives $Q'' = 1$. For $\ell \in \gamma_2(Q)$, $y \in Q$ and $c = [\ell, y]$, the element $[c, y]$ is central, so $[\ell, y^p] = c^p[c,y]^{\binom{p}{2}} = c^p$; and $y^p \in \gamma_2(Q)$ (its image lies in $\gamma_2(P)$, and $z \in \gamma_2(Q)$), so $[\ell, y^p] = 1$. Since $\gamma_3(Q) \le \gamma_2(Q)$ is abelian and generated by such c, it has exponent p, and its order is $p\,|\gamma_3(P)| = p^2$.

Conversely, let Q be as in the statement. $\gamma_4(Q)$ maps into $\gamma_4(P) = 1$, so $\gamma_4(Q) = \langle z\rangle$. The image of $P_4(Q) \ge \gamma_4(Q)$ in P is $P_4(P) = 1$, so $P_4(Q) = \langle z\rangle$ and $P_5(Q) = [\langle z\rangle, Q]\langle z\rangle^p = 1$. As $z \in \gamma_2(Q) \le \Phi(Q)$, $d(Q) = d(P)$, so Q is an immediate descendant of P [OBr90]. □

Step 2: (N1)–(N3) are necessary. Let Q be as in Lemma C.1 and $L = L(Q)$: $L_1 = V$, $L_2 \cong \gamma_2(P)/\gamma_3(P)$, $L_3 \cong \gamma_3(P) = \langle w\rangle$, $L_4 = \langle z\rangle$. Write $[\ell, x] = \varphi_\ell(x)w$ in L_3 for $\ell \in L_2$, $x \in L_1$, and $[w, x] = \psi(x)z$. The Jacobi identity for $\ell \in L_2$ and $x, y \in L_1$ gives $[\ell, [x,y]] = \varphi_\ell(x)\psi(y) - \varphi_\ell(y)\psi(x)$, and the left side is 0 because $Q'' = 1$. Moreover $\psi \neq 0$ because $\gamma_4(Q) = [\gamma_3(Q), Q] \neq 1$. So every φ_ℓ is a multiple of ψ, and the map $\ell \mapsto \varphi_\ell$ has rank 1 (it is nonzero since $\gamma_3(P) \neq 1$). Its kernel is spanned by the class of some $v \in \gamma_2(P)\backslash\gamma_3(P)$, and $[v, x] = w^{\varphi_v(x)} = 1$ for all $x \in P$. Thus $Z(P) \cap \gamma_2(P) \supseteq \langle v, w\rangle = \langle v\rangle\gamma_3(P)$, which has order p^2; equality holds because $\gamma_2(P) \not\le Z(P)$ (as $\gamma_3(P) \neq 1$) and $|\gamma_2(P)| = p^3$. This is (N1). If $u \in \gamma_2(P) \setminus Z(P)$ and $f = \varphi_u$, then $\psi = cf$ with $c \neq 0$. (N2): for $\ell \in \gamma_2(Q)$ and $y \in Q$, $[\ell^p, y] = [\ell, y]^p = 1$, so $\ell^p \in Z(Q) \cap P_3(Q) = Z(Q) \cap \gamma_3(Q)$. But $\gamma_3(Q) = \langle \tilde{w}\rangle \times \langle z\rangle$

for a preimage $\tilde{w}$ of w, and $[\tilde{w}^a z^b, x] = z^{a\psi(x)}$, so $Z(Q) \cap \gamma_3(Q) = \langle z \rangle$. Hence $\gamma_2(P)$ has exponent p; it is abelian since $P'' \le \gamma_4(P) = 1$. (N3): for $x, y \in P$ and $a = [y, x] \in \gamma_2(P)$, $[y, x^p] = a^p [a, x]^{\binom{p}{2}} [a, x, x]^{\binom{p}{3}} = 1$ by (N2), $|\gamma_3(P)| = p$ and $\gamma_4(P) = 1$.

Step 3: normal form. Assume (N1)–(N3) and choose u, v, w, x_i as in the theorem. Then $\gamma_2(P) = \langle u \rangle \times \langle v \rangle \times \langle w \rangle$ is elementary of order p^3, $Z(P) \cap \gamma_2(P) = \langle v, w \rangle \ni x_i^p$, $[u, x_i] = w^{f_i}$, $[v, x_i] = [w, x_i] = 1$, $\omega_r(j, i) = -\omega_r(i, j)$, and $f \neq 0$ since $\gamma_3 = [\gamma_2, P] \neq 1$. These relations, with all relative orders p, form a consistent pc-presentation of P on $x_1, \dots, x_d, u, v, w$, since $|P| = p^{d+3}$.

Step 4: (N4) is necessary. Let Q be as in Lemma C.1, and keep the names x_i, u, v, w for fixed preimages. The group $A = \gamma_2(Q) = \langle u, v, w, z \rangle$ is abelian; we write it additively and put $D_x a = [a, x] = a^x - a$ for $x \in Q$. Each D_x is an endomorphism of A. We have $D_x D_y = D_y D_x$, since $xy = yx[x, y]$ and $[x, y] \in A$ acts trivially on A. Every product of three D's vanishes, since $[A, Q, Q, Q] \le \gamma_5(Q) = 1$. For some $\tau_{ij}, \rho_i, \eta_i, \sigma_i, \mu_u, \mu_v \in \mathbb{F}_p$,

$$[x_i, x_j] = \omega_1(i, j)u + \omega_2(i, j)v + \omega_3(i, j)w + \tau_{ij} z, \qquad x_i^p = \alpha_i v + \beta_i w + \rho_i z,$$

$$D_{x_i} u = f_i w + \sigma_i z, \quad D_{x_i} v = \eta_i z, \quad D_{x_i} w = c f_i z,$$

$$pu = \mu_u z, \quad pv = \mu_v z, \quad pw = 0,$$

by Step 2 and Lemma C.1.

(J): put $a = [x, y] \in A$ in the Hall–Witt identity

$$[x, y^{-1}, t]^y \, [y, t^{-1}, x]^t \, [t, x^{-1}, y]^x = 1.$$

Then $[x, y^{-1}] = (a^{-1})^{y^{-1}} = -(1 - D_y + D_y^2)a$, and, dropping products of three D's,

$$[x, y^{-1}, t]^y = (1 + D_y) D_t (-a + D_y a) = -D_t a = -[x, y, t].$$

So $[x, y, t] + [y, t, x] + [t, x, y] = 0$ holds exactly in Q. For $(x, y, t) = (x_i, x_j, x_k)$,

$$[x_i, x_j, x_k] = D_{x_k}[x_i, x_j] = \omega_1(i, j)(f_k w + \sigma_k z) + \big(\omega_2(i, j)\eta_k + c\,\omega_3(i, j) f_k\big) z.$$

The w-component of the cyclic sum vanishes already in P; the z-component is (J). (E): let $x = x_j$, $y = x_i$ and $a = [x, y]$. On the one hand $(x^p)^y = x^p + D_y x^p = x^p + (\alpha_j \eta_i + c\beta_j f_i) z$. On the other hand

$$(x^p)^y = (xa)^p = x^p + \sum_{m=0}^{p-1} (1 + D_x)^m a = x^p + pa + \tbinom{p}{2} D_x a + \tbinom{p}{3} D_x^2 a.$$

Here $D_x a \in \gamma_3(Q)$ has exponent p, $pa = (\mu_u \omega_1(j, i) + \mu_v \omega_2(j, i)) z$, $D_x^2 a = c\, f_j^2\, \omega_1(j, i) z$, and $\binom{p}{3} \equiv \varepsilon_p \pmod p$. Comparing and using $\omega_r(j, i) = -\omega_r(i, j)$ gives (E) (for $i = j$, $a = 0$ and (E) says $[x_j^p, x_j] = 1$). Dividing by c gives a solution with $c = 1$.

Step 5: (N4) is sufficient. Take a solution (with any $c \neq 0$) and define a pc-presentation $\mathcal{Q}$ on $x_1, \dots, x_d, u, v, w, z$, in this order, all relative orders p:

$$x_i^p = v^{\alpha_i} w^{\beta_i}, \quad u^p = z^{\mu_u}, \quad v^p = z^{\mu_v}, \quad w^p = z^p = 1,$$
$$x_j^{x_i} = x_j\, u^{\omega_1(j,i)} v^{\omega_2(j,i)} w^{\omega_3(j,i)} \ (j > i),$$
$$u^{x_i} = u\, w^{f_i} z^{\sigma_i}, \quad v^{x_i} = v\, z^{\eta_i}, \quad w^{x_i} = w\, z^{c f_i},$$

and all other pairs of generators commute. Modulo z this is the presentation of Step 3. A pc-presentation is consistent if and only if the following test words collect to the same normal form [Sim94, Ch. 9], [VL84]: for generators $g_k > g_j > g_i$, $(g_k g_j) g_i = g_k (g_j g_i)$, $(g_j^p) g_i = g_j^{p-1} (g_j g_i)$, $g_j (g_i^p) = (g_j g_i) g_i^{p-1}$, and $(g_i^p) g_i = g_i (g_i^p)$. All of them hold modulo z, so each test compares exponents of z, computed by collection from the defining relations alone. Let $A = \langle u, v, w, z \rangle$, which is abelian by the relations. Conjugation by x_i, as the relations give it on u, v, w, z, maps them into A and respects the power relations of A (this is the content of the tests $(\ell^p) x_i$, checked below), so it induces an endomorphism $1 + D_i$ of A. Let a_{kj} be the right-hand word for $[x_k, x_j]$, $k > j$. Then

$$(x_k x_j) x_i = x_i x_j x_k \big[(1 + D_k) a_{ji} + a_{ki} + (1 + D_i) a_{kj} \big],$$
$$x_k (x_j x_i) = x_i x_j x_k \big[a_{kj} + (1 + D_j) a_{ki} + a_{ji} \big],$$

and the difference is $D_k a_{ji} + D_i a_{kj} - D_j a_{ki} = -J(i,j,k)\, z$, its w-component vanishing by the identity in P. The power tests repeat the computation of $(x_j a)^p$ in Step 4, which uses only the relations: $(x_j^p) x_i$ and $x_j (x_i^p)$ $(j > i)$ give $\pm E(i,j) z$ and $\pm E(j,i) z$, and $(x_i^p) x_i$ gives $E(i,i) z$. For $\ell \in \{u, v, w\}$, the test (ℓ, x_j, x_i) is the Jacobi identity with ℓ: it gives $(f_j\, c f_i - f_i\, c f_j) z = 0$ for $\ell = u$, and 0 for $\ell = v, w$. The tests $(\ell^p) x_i$ and $\ell (x_i^p)$ hold because $p\, D_i \ell = 0$ and $(1 + D_i)^p \ell = \ell$, as $pw = pz = 0$, $D_i^3 = 0$ and $p \mid \binom{p}{2}$. The remaining tests, (ℓ, ℓ', x_i) with $\ell, \ell' \in \{u, v, w, z\}$ and those involving z, hold trivially: the images of u, v, w, z under conjugation by x_i lie in the abelian group A, and z is central. So $\mathcal{Q}$ is consistent; it defines a group Q of order p^{d+4} with z central and $Q/\langle z \rangle \cong P$. Choose k with $f_k \neq 0$. Since $u \in \gamma_2(Q) \langle z \rangle$, we get $w^{f_k} z^{\sigma_k} = [u, x_k] \in \gamma_3(Q)$, so $w \in \gamma_3(Q) \langle z \rangle$ and $1 \neq z^{c f_k} = [w, x_k] \in \gamma_4(Q)$. By Lemma C.1, P is nonterminal. □

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